

Introduction to $\mathbb{P}_{\max}$

Obrad Kasum

February 1, 2022

Abstract

We present basics of $\mathbb{P}_{\max}$, with the goal of proving that if $V = L(\mathbb{R}) \models \text{AD}$, then $V^{\mathbb{P}_{\max}} \models \text{ZFC} + \ulcorner \text{NS}_{\omega_1} \text{ is saturated} \urcorner$. Our sources are [Larson, 2010] and [Woodin, 2013].

Contents

1	Poset $\mathbb{P}_{\max}$	2
2	Existence of Conditions	6
3	Countable Closure	12
3.1	Curious Sequences	12
3.2	Proof of the Countable Closure	15
4	Subsets of ω_1 inside $L(\mathbb{R})^{\mathbb{P}_{\max}}$	19
4.1	Some Facts from Descriptive Set Theory	19
4.2	A Stronger Iterability Condition	21
4.3	The Generic Power Set of ω_1	25
5	Properties of $L(\mathbb{R})^{\mathbb{P}_{\max}}$	32

1 Poset $\mathbb{P}_{\max}$

I. **Definition.** Let M be a transitive model for ZFC^- , let $M \models \ulcorner I \urcorner$ is a precipitous ideal on $\omega_1^\ulcorner$, and let $\gamma \leq \text{Ord}$. A *putative generic γ -iteration of (M, I)* is a family $(g_\alpha : \alpha < \gamma)$ satisfying the following, for the uniquely determined $(M_\alpha, j_{\alpha, \beta} : \alpha \leq \beta < \gamma)$.

- a. $j_{\alpha, \beta} : M_\alpha \prec M_\beta$ form a commutative system of elementary embeddings, with $j_{\alpha, \alpha} = \text{id}$.
- b. $M_0 := M$
- c. For $\alpha + 1 < \gamma$, M_α is well-founded and g_α is M_α -generic for $j_{0, \alpha}(I)$.
- d. For $\alpha + 1 < \gamma$, $M_{\alpha+1} = \text{Ult}(M_\alpha, g_\alpha)$ and $j_{\alpha, \alpha+1}$ is the ultrapower embedding.
- e. For a limit $\delta < \gamma$, M_δ is the direct limit of the system $(M_\alpha, j_{\alpha, \beta} : \alpha \leq \beta < \delta)$ and $j_{\alpha, \delta}$ are the direct limit embeddings.

II. **Remark.** Note that only the last model of a putative iteration (if there is one) can be ill-founded. $(M_\alpha)_{\alpha < \gamma}$ and $(j_{\alpha, \beta})_{\alpha \leq \beta < \gamma}$ are referred to as *models* and *embeddings* of the generic iteration. We sometimes use the term “putative generic iteration” for the family $(M_\alpha, j_{\alpha, \beta} : \alpha \leq \beta < \gamma)$. If there is the last model $M_{\gamma'}$ of this iteration, we use the term “putative generic iteration” also for the embedding $j_{0, \gamma'} : (M, I) \prec (M_{\gamma'}, j_{0, \gamma'}(I))$.

III. **Definition.** A putative generic iteration is well-founded if all of its models are well-founded¹. A *generic γ -iteration of (M, I)* is a putative generic γ -iteration of (M, I) which is well-founded.

IV. **Definition.** (M, I) is *generically γ -iterable* if every putative generic γ -iteration of (M, I) is well-founded.

V. **Remark.** (V, I) is generically Ord -iterable (in V). The reason why this is true (and irrelevant) is because there are no V -generic filters over V . We see that the relevance of this notion is dependent on existence of appropriate generic filters. The main case that we are interested is the case of a countable transitive model for ZFC. In this, we will use term “*generically iterable*” to refer to the case of generic ω_1^V -iterability².

VI. **Definition.** A *pre-condition* (for $\mathbb{P}_{\max}$) is a pair (M, I) , where:

- a. M is a countable transitive model for ZFC;
- b. $M \models \ulcorner I \urcorner$ is a normal ideal on $\omega_1^\ulcorner$;
- c. (M, I) is generically iterable.

VII. **Definition.** Conditions of $\mathbb{P}_{\max}$ are structures $(M, \in, I, a)$ satisfying:

¹Rr equivalently, the last model is well-founded, if there is one.

²Or equivalently, generic $\omega_1^V + 1$ -iterability.

- a. (M, I) is a pre-condition;
- b. $M \models \mathbf{MA}_{\omega_1}$
- c. $a \subseteq \omega_1^M$
- d. $M \models \omega_1 = \omega_1^{L[a]}$.

VIII. **Notation.** If $p = (M, \in, I, a)$ is a $\mathbb{P}_{\max}$ -condition, then we write $I^p := I$ and $a^p := a$.

IX. **Proposition.** Let $p \in \mathbb{P}_{\max}$ and let $i : p \prec q$ be a generic iteration of p of countable length. Then $(q, \in, i(I^p), i(a^p)) \in \mathbb{P}_{\max}$.

Proof.

- 1° Only the requirement of generic iterability needs a comment.
- 2° Any putative generic iteration $j : q \prec q'$ yields a putative generic iteration $j \circ i : p \prec q'$.
- 3° Since p is generically iterable, q' is well-founded.
- 4° This means that j is a generic iteration.
- 5° Thus, q is generically iterable.

□

X. **Proposition.** Let $p \in \mathbb{P}_{\max}$. Then $\omega_1^p = \omega_1^{L[a^p]}$ and a^p is unbounded in ω_1^p .

Proof.

- 1° Let $i : p \prec p_1$ be a generic ultrapower. Then p and p_1 are of the same height, so $\omega_1^p = \omega_1^{L[a^p]^p} = \omega_1^{L[a^p]^{p_1}}$.
- 2° There is a generic ω_1 -iteration $j : p_1 \prec p_{\omega_1}$.
- 3° Since

$$p_1 \models \omega_1^p = \omega_1^{L[a^p]}$$

and since ω_1^p and a^p are fixed under j , we conclude

$$p_{\omega_1} \models \omega_1^p = \omega_1^{L[a^p]}.$$

- 4° This means $\omega_1^p = \omega_1^{L[a^p]^{p_{\omega_1}}}$.
- 5° Since $p_{\omega_1} \cap \text{Ord} > \omega_1^V$, we conclude by condensation that $\omega_1^{L[a^p]} = \omega_1^{L[a^p]^{p_{\omega_1}}} = \omega_1^p$.
- 6° To show that a^p is unbounded in ω_1^p , assume otherwise.
- 7° Let $i : p \prec q$ be a generic ultrapower. Then $a^q = i(a^p) = a^p$.
- 8° $\omega_1^q = \omega_1^{L[a^q]^q} = \omega_1^{L[a^q]} = \omega_1^{L[a^p]} = \omega_1^{L[a^p]^p} = \omega_1^p$, which is a contradiction.

□

XI. **Proposition.** Suppose that MA_{ω_1} holds. Let $(x_\alpha : \alpha < \omega_1)$ be an almost disjoint family of subsets of ω . Then for every $A \subseteq \omega_1$, there is $y \subseteq \omega$ such that

$$(\forall \alpha < \omega_1)(\alpha \in A \iff |y \cap x_\alpha| = \omega).$$

Proof. Lemma 2.17 of [Kunen, 2014]. □

XII. **Proposition.** Let $p \in \mathbb{P}_{\max}$ and $A \subseteq \delta \leq \omega_1$. Then there is a most one generic iteration $i : p \prec p^*$ such that

$$i(\omega_1^p) = \delta, \quad i(a^p) = A.$$

Proof.

- 1° Let $(g_\alpha : \alpha \leq \gamma)$ and $(g'_\alpha : \alpha \leq \gamma')$ be two such iterations, with $\gamma \leq \gamma'$. Let $(p_\alpha, i_{\alpha, \beta})_{\alpha \leq \beta}$ and $(p'_\alpha, i'_{\alpha, \beta})_{\alpha \leq \beta}$ be the models and embeddings of these iterations.
- 2° Assume inductively that $g_\beta = g'_\beta$ for $\beta < \alpha$ and let us show that $g_\alpha = g'_\alpha$, where $\alpha < \gamma$.
- 3° We have $p_\alpha = p'_\alpha$.
- 4° Let $\text{code} : \omega \times \omega \rightarrow \omega$ be the standard bijection.
- 5° We define f such that $L[a^p]^p \models f : \omega_1 \rightarrow P(\omega)$ by recursion.
- 6° For $n < \omega$, we set $f(n) := \{\text{code}(n, i) : i < \omega\}$.
- 7° For $\alpha \in [\omega, \omega_1^p]$, assume recursively that $f \restriction \alpha$ is defined and almost disjoint. Then $f[\alpha]$ is not maximal. Let $f(\alpha)$ be the $<_{L[a^p]^p}$ -minimal subset of ω almost disjoint from every element of $f[\alpha]$.
- 8° By elementarity, inside $L[A]^{p_\gamma}$, we have:
 - a. $i_{0, \gamma}(f) : \omega_1 \rightarrow P(\omega)$
 - b. for $n < \omega$, we set $i_{0, \gamma}(f) := \{\text{code}(n, i) : i < \omega\}$;
 - c. $i_{0, \gamma}(f)$ is the $<_{L[A]}$ -minimal subset of ω almost disjoint from every element of $i_{0, \gamma}(f)[\alpha]$.
- 9° Since $\omega_1^{L[A]^{p_\gamma}} = \omega_1^{L[A]}$, we have inside $L[A]$:
 - a. $i_{0, \gamma}(f) : \omega_1 \rightarrow P(\omega)$
 - b. for $n < \omega$, we set $i_{0, \gamma}(f) := \{\text{code}(n, i) : i < \omega\}$;
 - c. $i_{0, \gamma}(f)$ is the $<_{L[A]}$ -minimal subset of ω almost disjoint from every element of $i_{0, \gamma}(f)[\alpha]$.
- 10° Analogously, inside $L[A]$:
 - a. $i'_{0, \gamma'}(f) : \omega_1 \rightarrow P(\omega)$
 - b. for $n < \omega$, we set $i'_{0, \gamma'}(f) := \{\text{code}(n, i) : i < \omega\}$;

c. $i'_{0,\gamma'}(f)$ is the $<_{L[A]}$ -minimal subset of ω almost disjoint from every element of $i'_{0,\gamma'}(f)[\alpha]$.

11° We conclude $i_{0,\gamma}(f) = i'_{0,\gamma'}(f)$.

12° Since $p_\alpha \models \mathbf{MA}_{\omega_1}$, for every $S \in P(\omega_1)^{p_\alpha}$, there is $x_S \in P(\omega)^{p_\alpha}$ such that

$$(\forall \xi < \omega_1^{p_\alpha})(\xi \in S \iff |i_{0,\alpha}(f)(\xi) \cap x_S| = \omega).$$

(Cf. 1-XI.)

13° By elementarity, for every $S \in P(\omega_1)^{p_\alpha}$,

$$(\forall \xi < \omega_1^{p_\gamma})(\xi \in i_{\alpha,\gamma}(S) \iff |i_{0,\gamma}(f)(\xi) \cap x_S| = \omega),$$

$$(\forall \xi < \omega_1^{p'_{\gamma'}})(\xi \in i'_{\alpha,\gamma'}(S) \iff |i'_{0,\gamma'}(f)(\xi) \cap x_S| = \omega).$$

14° For every $S \in P(\omega_1)^{p_\alpha}$:

$$S \in g_\alpha \iff \omega_1^{p_\alpha} \in i_{\alpha,\gamma}(S) \iff |i_{0,\gamma}(f)(\omega_1^{p_\alpha}) \cap x_S| = \omega \iff$$

$$|i'_{0,\gamma'}(f)(\omega_1^{p_\alpha}) \cap x_S| = \omega \iff \omega_1^{p_\alpha} \in i'_{\alpha,\gamma'}(S) \iff S \in g'_\alpha.$$

15° Thus, $g_\alpha = g'_\alpha$, concluding the inductive argument.

16° We have established that the $(g_\alpha)_\alpha$ is an initial part of $(g'_\alpha)_\alpha$. Since these iterations both map ω_1^p to δ , they must be equal.

□

XIII. Let $p, q \in \mathbb{P}_{\max}$. By the previous proposition, there can be at most one iteration i of p such that

$$i(\omega_1^p) = \omega_1^q, \quad i(I^p) = I^q, \quad i(a^p) = a^q.$$

When it exists, we call this iteration *the iteration of p given by q* .

XIV. **Definition.** For $p, q \in \mathbb{P}_{\max}$, we define $q < p$ to hold exactly in the case when there exists the iteration of p given by q and it belongs to q .

XV. **Proposition.** $<$ is a strict order.

Proof.

1° We only need to establish transitivity.

2° Suppose that $r < q < p$ and let us show $r < p$.

3° Let i be the iteration of p given by q ; we have $i \in q$.

4° Let j be the iteration of q given by r ; we have $j \in r$.

5° $j(i)$ is the iteration of p given by r ; obviously, $j(i) \in r$.

□

2 Existence of Conditions

- I. **Lemma.** Suppose that there is a putative generic iteration of an inner model W that is not well-founded. Then there is some η such that $W^{\text{Col}(\omega, \eta)} \models \ulcorner \text{there is a putative generic iteration of } W \text{ that is not well-founded} \urcorner$.

Proof.

- 1° There is $\kappa > (2^{2^{\omega_1}})^W$ a regular cardinal of W such that there is a putative generic γ -iteration $j : H(\kappa)^W \prec H_\gamma$ such that H_γ is ill-founded.
- 2° Let $\eta := \max\{2^\kappa, \gamma\}$ and let $\Gamma \rightsquigarrow V^{\text{Col}(\omega, \eta)}$.
- 3° In $V[\Gamma]$ and in $W[\Gamma]$, 1° is a Σ_1^1 -statement about reals (note that the regularity of κ in W is in fact expressible in $H(\kappa^+)^W$).
- 4° 1° is true in $V[\Gamma]$, so Σ_1^1 -absoluteness implies that it is true in $W[\Gamma]$.
- 5° This means that it is forced by $\emptyset$ in $\text{Col}(\omega, \eta)$ over W .
- 6° Thus, η is as required.

□

- II. **Lemma.** Suppose that $M \models \ulcorner I \text{ is a normal ideal on } \omega_1 \urcorner$ and let $(M_\alpha, j_{\alpha, \beta} : \alpha \leq \beta < \gamma + 1)$ be a putative generic iteration of (M, I) . Then

$$M_\gamma = \{j_{0, \gamma}(f)(\omega_1^{M_{i_0}} \dots \omega_1^{M_{i_{n-1}}}) : n < \omega, f : (\omega_1^M)^n \rightarrow M, f \in M, \\ i_0 < \dots < i_{n-1} < \gamma\}.$$

Proof. This follows easily by induction. □

- III. **Proposition.** Suppose that I is a normal precipitous ideal on ω_1 . Then (V, I) is Ord-iterable in every generic extension of V .

Proof.

- 1° Assume otherwise. Then for some η , $\|\ulcorner \text{there is a putative generic iteration of } V \text{ that is ill-founded} \urcorner\|_{\text{Col}(\omega, \eta)} > 0$. The homogeneity implies that this norm is in fact 1.
- 2° This means that in $V^{\text{Col}(\omega, \eta)}$, there is a regular cardinal κ of V such that there is a putative generic iteration of $H(\kappa)^V$ that is ill-founded. Again by homogeneity, the value κ is decided by the weakest condition.
- 3° Let (γ, κ, θ) be the lexicographically minimal pair of ordinals such that:
 - a. $\kappa > 2^{2^{\omega_1}}$ is a regular cardinal;
 - b. $\theta < \kappa$
 - c. there is η such that $V^{\text{Col}(\omega, \eta)} \models \ulcorner \text{there is a putative generic } (\gamma + 1)\text{-iteration } j : (H(\kappa)^V, I) \prec (M_\gamma, I_\gamma) \text{ such that } j(\theta) \text{ is ill-founded} \urcorner$.

- 4° Fix such an η with $\eta > \max\{2^\kappa, \gamma\}$. Let $\zeta > \eta$ be a beth fixed point of cofinality $> \eta$ and let $\Gamma \rightsquigarrow V^{\text{Col}(\omega, \zeta)}$.
- 5° In $V[\Gamma \cap \text{Col}(\omega, \eta)]$, let $j_{0, \gamma} : H(\kappa)^V \prec M_\gamma$ be a generic $(\gamma + 1)$ -iteration such that $j_{0, \gamma}(\theta)$ is ill-founded.
- 6° γ is a limit ordinal.
- 7° Thus, there are $\gamma' < \gamma$ and $\theta' < j_{0, \gamma'}(\theta)$ such that $j_{\gamma', \gamma}(\theta')$ is ill-founded.
- 8° Putative iteration $j_{0, \gamma}$ produces a putative iteration $k_{0, \gamma}$ of V . All the models except the last one are indeed well-founded by the choice of (γ, κ, θ) . Denote the models of this iteration by N_α ($\alpha < \gamma + 1$).
- 9° $V[\Gamma] \models \ulcorner \text{there is a putative generic } (\gamma + 1 - \gamma')\text{-iteration of } (H(k_{0, \gamma'}(\kappa))^{N_{\gamma'}}, k_{0, \gamma'}(I)) \text{ such that the image of } \theta' \text{ is ill-founded} \urcorner$.
- 10° By the previous lemma and the choice of ζ , we see that ζ remains fixed throughout the iteration $k_{0, \gamma}$. In particular, in $N_{\gamma'}$, ζ is a beth fixed point above $k_{0, \gamma'}(\eta)$.
- 11° This means that there is a generic $\Delta \rightsquigarrow N_{\gamma'}^{\text{Col}(\omega, k_{0, \gamma'}(\eta))}$.
- 12° From 9° by Σ_1^1 -absoluteness, it follows that $N_{\gamma'}[\Delta] \models \ulcorner \text{there is a putative generic } (\gamma + 1 - \gamma')\text{-iteration of } (H(k_{0, \gamma'}(\kappa))^{N_{\gamma'}}, k_{0, \gamma'}(I)) \text{ such that the image of } \theta' \text{ is ill-founded} \urcorner$.
- 13° From 3° by elementarity of $k_{0, \gamma'}$, it follows that in $N_{\gamma'}^{\text{Col}(\omega, k_{0, \gamma'}(\eta))}$, the triple

$$(k_{0, \gamma'}(\gamma), k_{0, \gamma'}(\kappa), k_{0, \gamma'}(\theta))$$

is the lexicographically minimal one for which there exists a putative generic $(k_{0, \gamma'}(\gamma) + 1)$ -iteration $j : (H(k_{0, \gamma'}(\kappa))^{N_\alpha}, k_{0, \gamma'}(I)) \prec (M^*, I^*)$ such that $j(k_{0, \gamma'}(\theta))$ is ill-founded.

- 14° Thus,

$$(k_{0, \gamma'}(\gamma), k_{0, \gamma'}(\kappa), k_{0, \gamma'}(\theta)) \leq (\gamma - \gamma', k_{0, \gamma'}(\kappa), \theta').$$

- 15° Since $k_{0, \gamma'}(\gamma) \geq \gamma - \gamma'$, the first two coordinates are equal, implying $j_{0, \gamma'}(\theta) \leq \theta'$. This is a contradiction.

□

IV. **Theorem.** Suppose that W is an inner model and suppose that $W \models \ulcorner I \text{ is a normal precipitous ideal on } \omega_1 \urcorner$. Then (W, I) is generically **Ord**-iterable.

Proof. Immediately from 2-I and 2-III. □

V. **Theorem.** Suppose that W is an inner model, that $\lambda < \omega_1^V$ is an inaccessible cardinal of W , and that $W \models \ulcorner I \text{ is a normal precipitous ideal on } \omega_1 \urcorner$. Then $(W \upharpoonright \lambda, I)$ is a pre-condition for $\mathbb{P}_{\max}$.

Proof. We need to establish generic iterability. If it failed, then (W, I) would not be generically ω_1^V -iterable, which is a contradiction with the previous theorem. $\square$

VI. Proposition. Let I be a normal precipitous ideal on ω_1 . Suppose that $\mathbb{P}$ is a c.c.c. poset and $G \rightsquigarrow V^{\mathbb{P}}$. Then the ideal in $V[G]$ generated by I is normal and precipitous.

Proof.

- 1° Let I^G denote the ideal generated by I in $V[G]$. We will show that it is normal. For precipitousness, see Theorem 1 of [Kakuda, 1981].
- 2° Let $X_\alpha \in I^G$ for $\alpha < \omega_1$. We want to show $\nabla_{\alpha < \omega_1} X_\alpha \in I^G$.
- 3° There are $Y_\alpha \in I$ with $X_\alpha \subseteq Y_\alpha$.
- 4° In V , there is $\mathcal{Z} : \omega_1 \rightarrow [I]^\omega$ such that in $V[G]$, for every $\alpha < \omega_1$, we have $Y_\alpha \in \mathcal{Z}(\alpha)$.
- 5° Since I is uniform and normal, it is σ -complete.
- 6° Since I is σ -complete, we have $W_\alpha := \bigcup \mathcal{Z}(\alpha) \in I$ for every $\alpha < \omega_1$.
- 7° $\nabla_{\alpha < \omega_1} X_\alpha \subseteq \nabla_{\alpha < \omega_1} W_\alpha \in I$ and consequently, $\nabla_{\alpha < \omega_1} X_\alpha \in I^G$.

$\square$

VII. Proposition. Let $x \in \mathbb{R}$ and suppose that $x^\dagger$ exists. Then there is a pre-condition (M, I) such that $x \in M$ and $M \models \mathbf{MA}_{\omega_1}$.

Proof.

- 1° There is an inner model W such that:
 - a. $x \in W$
 - b. $W \models \mathbf{GCH}$
 - c. there is a measurable cardinal $\kappa < \omega_1^V$ in W ;
 - d. there is an inaccessible cardinal $\lambda \in (\kappa, \omega_1^V)$ in W .
- 2° Let $g \rightsquigarrow W^{\text{Col}(\omega, < \kappa)}$ be arbitrary. Then in $W[g]$ there is a normal precipitous ideal I on ω_1 .
- 3° λ remains inaccessible in $W[g]$ and $W[g] \models \mathbf{GCH}$.
- 4° There is a c.c.c. forcing $\mathbb{P}$ in $W[g] \restriction \lambda$ such that $W[g]^\mathbb{P} \models \mathbf{MA}_{\omega_1}$.
- 5° Let $G \rightsquigarrow W[g]^\mathbb{P}$. In $W[g][G]$, ordinal λ remains inaccessible and the ideal J generated by I is normal and precipitous (by 2-VI).
- 6° Let $M := W[g][G] \restriction \lambda$. By 2-V, (M, J) is generically iterable and thus, as required.

$\square$

VIII. **Theorem.** If $0^\dagger$ exists, then $\mathbb{P}_{\max} \neq \emptyset$.

Proof.

- 1° By 2-VII, there is a pre-condition (M, I) such that $M \models \mathbf{MA}_{\omega_1}$.
- 2° There is an $a \in M$ such that $a \subseteq \omega_1^M$ and $\omega_1^{L[a]^M} = \omega_1^M$.
- 3° $p := (M, \in, I, a)$ is a $\mathbb{P}_{\max}$ -condition.

□

IX. **Lemma.** Let (M, I) be a pre-condition and let $i : (M, I) \prec (M^*, I^*)$ be an ω_1 -iteration. Then $I^* \subseteq \mathbf{NS}_{\omega_1}$.

Proof.

- 1° The iterates are denoted by (M_ξ, I_ξ) , embeddings are denoted by $i_{\xi, \eta}$, and generics are denoted by g_ξ .
- 2° Let $X \in I^*$ and let us show $X \in \mathbf{NS}_{\omega_1}$.
- 3° There is $\alpha < \omega_1$ and $X' \in I_\alpha$ such that $X = i_{\alpha, \omega_1}(X')$.
- 4° $C := \{\omega_1^{M_\beta} : \alpha \leq \beta < \omega_1\}$ is club.
- 5° Since $X' \in I_\alpha$, we have $i_{\alpha, \beta}(X') \in I_\beta \subseteq g_\beta^{\mathbf{C}}$ for every $\beta \in [\alpha, \omega_1)$.
- 6° Thus, $\omega_1^{M_\beta} \in \omega_1^{M_{\beta+1}} - i_{\alpha, \beta+1}(X') \subseteq \omega_1 - X$ for every $\beta \in [\alpha, \omega_1)$.
- 7° This implies $X \cap C = \emptyset$, and consequently $X \in \mathbf{NS}_{\omega_1}$.

□

X. **Proposition.** Let (M, I) be a pre-condition and let J be a normal ideal on ω_1 . Then there is a generic ω_1 -iteration $i : (M, I) \prec (M_{\omega_1}, I_{\omega_1})$ such that $I_{\omega_1} = J \cap M_{\omega_1}$.

Proof.

- 1° Let $\{A_{i, \alpha} : i < \omega, \alpha < \omega_1\}$ be pairwise disjoint J -positive sets whose union is ω_1 .
- 2° We define a generic iteration $(g_\alpha : \alpha < \omega_1)$ of (M, I) by recursion on α . We also fix enumerations $\{B_i^\alpha : i < \omega\} = P(\omega_1)^{M_\alpha} - I_\alpha$.
- 3° Suppose recursively that $(g_\alpha : \alpha < \gamma)$ have been defined. In particular, iterate (M_γ, I_γ) is defined.
- 4° $\omega_1^{M_\gamma}$ belongs to exactly one $A_{i, \alpha}$.
- 5° Choose $g_\gamma \subseteq P(\omega_1)^{M_\gamma} - I_\gamma$ which is M_γ -generic and satisfies: if $\alpha \leq \gamma$, then $j_{\alpha, \gamma}(B_i^\alpha) \in g_\gamma$.
- 6° This concludes the recursion.

7° **Claim.** $I_{\omega_1} = J \cap M_{\omega_1}$

Proof.

- 1' By 2-IX, we have $I_{\omega_1} \subseteq \text{NS}_{\omega_1} \subseteq J$.
- 2' Let us show $(\supseteq)$. Let $X \in P(\omega_1)^{M_{\omega_1}} - I_{\omega_1}$ be arbitrary and let us show $X \in P(\omega_1) - J$
- 3' We have $X = j_{\alpha, \omega_1}(B_i^\alpha)$ for some $\alpha < \omega_1$ and some $i < \omega$.
- 4' $\{\omega_1^{M_\beta} : \beta \in [\alpha, \omega_1)\}$ is club and consequently, of J -measure one.
- 5' Since $A_{i, \alpha}$ is J -positive, the intersection S of these sets is J -positive.
- 6' For $\omega_1^{M_\beta} \in S$, we have $\omega_1^{M_\beta} \in A_{i, \alpha}$ and $\alpha \leq \beta$. Thus, $j_{\alpha, \beta}(B_i^\alpha) \in g_\beta$ and consequently, $\omega_1^{M_\beta} \in j_{\alpha, \beta+1}(B_i^\alpha) \subseteq X$.
- 7' We conclude $S \subseteq X$.
- 8' Since S is J -positive, we have $X \in P(\omega_1)^{M_{\omega_1}} - J$.

□

8° Thus, $(g_\alpha : \alpha < \omega_1)$ is as required.

□

XI. **Theorem.** Suppose that $x^\dagger$ exists for every $x \in \mathbb{R}$. Then for every $x \in \mathbb{R}$, set $\{p \in \mathbb{P}_{\max} : x \in p\}$ is dense.

Proof.

- 1° Let $p_0 \in \mathbb{P}_{\max}$ be arbitrary. We want to find $p < p_0$ with $x \in p$.
- 2° Let $y \in \mathbb{R}$ code x and p_0 .
- 3° By 2-VII, there is a pre-condition (M, I) with $y \in M$ and $M \models \text{MA}_{\omega_1}$.
- 4° By 2-X inside M , there is a generic iteration $i : p_0 \prec p_1$ of length ω_1 such that $I^{p_1} = I \cap p_1$.
- 5° Thus, $p := (W[g][G] \upharpoonright \lambda, \in, J, a^{p_1}) \in \mathbb{P}_{\max}$ and $p < p_0$.
- 6° Since $y \in p$, we also have $x \in p$.

□

XII. **Theorem.** Suppose that $x^\dagger$ exists for every $x \in \mathbb{R}$. Then for every $p_0, p_1 \in \mathbb{P}_{\max}$, there are $q_0 < p_0$ and $q_1 < p_1$ such that $\mathbb{P}_{\max} \upharpoonright q_0 \cong \mathbb{P}_{\max} \upharpoonright q_1$. In particular, if formula $\phi(x)$ and $a \in V$ are such that there exists $p \in \mathbb{P}_{\max}$ with $p \Vdash \phi(\check{a})$, then $\Vdash \phi(\check{a})$.

Proof.

- 1° By 2-VII, there is a pre-condition (M, I) such that $p_0, p_1 \in H_{\omega_1}^M$ and $M \models \text{MA}_{\omega_1}$.

- 2° By 2-X, inside M , there are iterations $j_0 : p_0 \prec p'_0$ and $j_1 : p_1 \prec p'_1$ of length ω_1 such that $I^{p'_0} = I \cap p'_0$ and $I^{p'_1} = I \cap p'_1$.
- 3° Let $q_i := (M, \in, I, a^{p'_i}) < p_i$ ($i = 0, 1$). j_i is the iteration of p_i given by q_i .
- 4° Let $r < q_0$ be arbitrary and let k be the iteration of q_0 given by r .
- 5° We define $f(r) := (M^r, \in, I^r, k(a^{p'_1}))$. It is clear that $f(r) < q_1$.
- 6° After setting $f(q_0) := q_1$, we get a mapping $f : \mathbb{P}_{\max} \restriction q_0 \rightarrow \mathbb{P}_{\max} \restriction q_1$. It is easily seen that f is an isomorphism.
- 7° Let us now verify the “in particular” part. Suppose $p \Vdash \phi(a)$, while $\nVdash \phi(a)$.
- 8° There is p_1 such that $p_1 \Vdash \neg\phi(a)$.
- 9° There are $q_0 < p$, $q_1 < p_1$, and an isomorphism $f : \mathbb{P}_{\max} \restriction q_0 \rightarrow \mathbb{P}_{\max} \restriction q_1$.
- 10° We have $q_0 \Vdash_{\mathbb{P}_{\max}} \phi(a) \iff \Vdash_{\mathbb{P}_{\max} \restriction q_0} \phi(a) \iff \Vdash_{\mathbb{P}_{\max} \restriction q_1} \phi(a) \Vdash q_1 \Vdash_{\mathbb{P}_{\max}} \phi(a)$.
- 11° This is a contradiction: $q_0 \leq p \Vdash \phi(a)$, while $q_1 \leq p_1 \Vdash \neg\phi(a)$.

□

3 Countable Closure

3.1 Curious Sequences

I. **Definition.** A *curious sequence* is a sequence $\mathcal{M} = ((M_i, I_i) : i < \omega)$ such that:

- a. (M_i, I_i) is a pre-condition;
- b. $\omega_1^{\mathcal{M}} := \omega_1^{M_i} = \omega_1^{M_j}$ for $i, j < \omega$;
- c. $M_i \in M_j$ for $i < j < \omega$;
- d. $M_j \models |M_i| = \omega_1$ for $i < j < \omega$;
- e. $I_j \cap M_i = I_i$ for $i < j < \omega$.

We define

$$\mathbb{B}^{\mathcal{M}} := \bigcup_{i < \omega} P^{M_i}(\omega_1^{\mathcal{M}}) \subseteq P(\omega_1^{\mathcal{M}}),$$

$$I^{\mathcal{M}} := \bigcup_{i < \omega} P^{M_i}(\omega_1^{\mathcal{M}}) \subseteq \mathbb{B}^{\mathcal{M}}.$$

II. **Remark.** $\mathbb{B}^{\mathcal{M}}$ is a boolean algebra and $I^{\mathcal{M}}$ is an ideal in it.

III. **Definition.** An ultrafilter $G \subseteq \mathbb{B}^{\mathcal{M}}$ is said to be $\mathcal{M}$ -generic if $G \cap M_i \rightsquigarrow M_i^{I_i}$ for every $i < \omega$.

IV. **Definition.** Let $\mathcal{M}$ be a curious sequence and let G be $\mathcal{M}$ -generic. The *ultrapower* $\text{Ult}(\mathcal{M}, G)$ is a curious sequence whose i^{th} model is equal to

$$\left((\omega_1^{\mathcal{M}} \rightarrow M_i) \cap \bigcup_{i < \omega} M_i \right) / G.$$

V. **Remark.** It is easily verifiable that this is indeed a curious sequence.

VI. **Proposition.** Let $\mathcal{M}$ be a curious sequence and let G be $\mathcal{M}$ -generic. For every formula $\phi(x \dots)$ and every $i < \omega$, we have

$$\text{Ult}(\mathcal{M}, G)_i \models \phi([f] \dots) \iff \{ \alpha < \omega_1^{\mathcal{M}} : \mathcal{M}_i \models \phi(f(\alpha) \dots) \} \in G.$$

VII. **Remark.** The previous proposition (i.e. Łoś's theorem) ensures that the ultrapower embeddings are elementary.

VIII. **Definition.** The *ultrapower embeddings* $j_G^i : \mathcal{M}_i \prec \text{Ult}(\mathcal{M}, G)_i$ are defined as to map an element x to the class of the constant function with value x .

IX. **Remark.** It is easily verifiable that

$$\text{Ult}(\mathcal{M}, G)_i = j^k(\mathcal{M}_i)$$

for $k > i$.

X. **Remark.** Let $\mathcal{M}$ be a curious sequence and let G be $\mathcal{M}$ -generic. Then

$$\omega_1^{\mathcal{M}} = [\text{id}_{\omega_1}]_G \in \text{Ult}(\mathcal{M}, G)_i$$

(for $i < \omega$).

XI. **Lemma.** Let $\mathcal{M}$ be a curious sequence and suppose that for every $i < \omega$, there is $x_i \in \mathbb{R} \cap \mathcal{M}_{i+1}$ such that

- a. $x_i^\# \in \mathcal{M}_{i+1}$
- b. the $(\omega_1^{\mathcal{M}} + 1)^{\text{th}}$ x_i -indiscernible is $\geq \text{Ord}^{\mathcal{M}_i}$
- c. if $\mathcal{M}_i \models \ulcorner C \urcorner$ is a club in $\omega_1^\top$, then C contains all x_i -indiscernibles below $\omega_1^{\mathcal{M}}$ starting from some.

Let G be $\mathcal{M}$ -generic. Then

$$\omega_1^{\text{Ult}(\mathcal{M}, G)} = \bigcup_{i < \omega} \text{Ord}^{\mathcal{M}_i}.$$

Proof.

- 1° Let j denote the ultrapower embedding.
- 2° For $i < \omega$, let $f_i : \omega_1^{\mathcal{M}} \rightarrow \omega_1^{\mathcal{M}}$ be defined by mapping ordinal α to the first x_i -indiscernible $> \alpha$.
- 3° Let us verify that for $h \in \mathcal{M}_i$, we have inside $\mathcal{M}_{i+1}$ that

$$\{\alpha < \omega_1^{\mathcal{M}} : h(\alpha) < f_i(\alpha)\}$$

contains a club.

- 4° Consider club $C := \{\alpha < \omega_1^{\mathcal{M}} : h[\alpha] \subseteq \alpha\} \in \mathcal{M}_i$. By [c](#), there is a x_i -indiscernible ι_0 such that for every x_i -indiscernible $\iota \geq \iota_0$, we have $\iota \in C$.
- 5° For $\iota \geq \iota_0$, let ι' be the next x_i -indiscernible. Since $\iota' \in C$, we have $h(\iota) < \iota' = f_i(\iota)$. This shows [3°](#).
- 6° By elementarity, $[f_i]_G = j^{i+1}(f_i)(\omega_1^{\mathcal{M}})$ is the first x_i -indiscernible $> \omega_1^{\mathcal{M}}$, in the sense of $\text{Ult}(\mathcal{M}, G)_{i+1}$. In other word, it is equal to the value of the critical point in the $(\omega_1^{\mathcal{M}} + 1)^{\text{st}}$ iterate of the mouse $x_i^\#$.
- 7° Since $\omega_1^{\mathcal{M}}$ is well-founded, this iterate has the same meaning in V and in $\text{Ult}(\mathcal{M}, G)_{i+1}$. In particular, $[f_i]_G$ is the first x_i -indiscernible above $\omega_1^{\mathcal{M}}$ in the sense of V .
- 8° Thus, $\omega_1^{\text{Ult}(\mathcal{M}, G)} = \sup\{[f_i]_G : i < \omega\} = \sup\{\text{Ord}^{\mathcal{M}_i} : i < \omega\}$. (The first equality follows from [3°](#), while the second one follows by [a](#), [b](#), [7°](#).)

□

XII. **Definition.** A *putative generic iteration* of a curious sequence $\mathcal{M}$ is determined by successively choose generic filters and constructing ultrapowers; at limit stages, we take direct limits. This iteration produces curious sequences $(\mathcal{M}_\alpha : \alpha < \gamma)$, where γ is the length of the iteration, together with embeddings

$$j_{\alpha,\beta}^i : (\mathcal{M}_\alpha)_i \prec (\mathcal{M}_\beta)_i,$$

for $i < \omega$ and $\alpha \leq \beta < \gamma$. All models of such an iteration are well-founded, except possibly the last one. *Generic iterations* are those whose all models are well-founded.

XIII. **Definition.** A curious sequence is *iterable* if its every putative generic iteration of length $\leq \omega_1 + 1$ is well-founded.

XIV. **Lemma.** Let $\mathcal{M}$ be a curious sequence and suppose that for every $i < \omega$, there is $x_i \in \mathbb{R} \cap \mathcal{M}_{i+1}$ such that

- a. $x_i^\# \in \mathcal{M}_{i+1}$
- b. the $(\omega_1^{\mathcal{M}} + 1)^{\text{th}}$ x_i -indiscernible is $\geq \text{Ord}^{\mathcal{M}_i}$
- c. if $\mathcal{M}_i \models \ulcorner C \text{ is a club in } \omega_1 \urcorner$, then C contains all x_i -indiscernibles below $\omega_1^{\mathcal{M}}$ starting from some.

Let $\mathcal{N}$ be an iterate of $\mathcal{M}$. Then $\omega_1^{\mathcal{N}}$ is well-founded.

Proof. This follows from 3.1-XI by induction. □

XV. **Proposition.** Let $\mathcal{M}$ be a curious sequence and suppose that for every $i < \omega$, there is $x_i \in \mathbb{R} \cap \mathcal{M}_{i+1}$ such that

- a. $x_i^\# \in \mathcal{M}_{i+1}$
- b. the $(\omega_1^{\mathcal{M}} + 1)^{\text{th}}$ x_i -indiscernible is $\geq \text{Ord}^{\mathcal{M}_i}$
- c. if $\mathcal{M}_i \models \ulcorner C \text{ is a club in } \omega_1 \urcorner$, then C contains all x_i -indiscernibles below $\omega_1^{\mathcal{M}}$ starting from some.

Then $\mathcal{M}$ is iterable.

Proof.

- 1° Let $j : \mathcal{M} \rightarrow \mathcal{N}$ be a countable length iteration of $\mathcal{M}$ and let $i < \omega$. We want to show that $(\text{Ord} \cap \mathcal{N}_i)^{\mathcal{N}_{i+1}}$ is well-founded.
- 2° By assumption, $\mathcal{M}_{i+1} \models \ulcorner \text{Ord} \cap \mathcal{M}_i \text{ is } \leq \text{than the } (\omega_1 + 1)^{\text{th}} x_i\text{-indiscernible} \urcorner$.
- 3° By elementarity of j^{i+1} , we have $\mathcal{N}_{i+1} \models \ulcorner \text{Ord} \cap \mathcal{N}_i \text{ is } \leq \text{than the } (\omega_1 + 1)^{\text{th}} x_i\text{-indiscernible} \urcorner$.
- 4° By 3.1-XIV, $\omega_1^{\mathcal{N}}$ is well-founded. Consequently, the $(\omega_1 + 1)^{\text{th}}$ x_i -indiscernible in the sense of $\mathcal{N}_{i+1}$ is the $(\omega_1 + 1)^{\text{th}}$ x_i -indiscernible in the sense of V . In particular, it is well-founded.

5° By 3°, we have that $(\text{Ord} \cap \mathcal{N}_i)^{\mathcal{N}_{i+1}}$ is well-founded.

□

XVI. **Proposition.** Suppose that $\mathcal{M}$ is an iterable curious sequence and let J be a normal ideal on ω_1 . Then there is a generic ω_1 -iteration $\mathcal{M} \rightarrow \mathcal{N}$ such that $I^{\mathcal{N}_i} = J \cap \mathcal{N}_i$ for every i .

Proof. This is a variation on the theme of lemma 2-X.

□

3.2 Proof of the Countable Closure

I. **Lemma.** If there is a precipitous ideal, then every real has a sharp.

Proof.

- 1° In some generic extension $V[G]$, there is an elementary embedding $j : V \prec M$ with $\text{crit } j = \omega_1^V$.
- 2° For $x \in \mathbb{R}^V$, we have $j \restriction L[x] : L[x] \prec L[x]$, so there is $x^\sharp$ in $V[G]$.
- 3° Since forcing cannot add a sharp, we have $x^\sharp \in V$.

□

II. **Lemma.** Every $\mathbb{P}_{\max}$ -condition is closed for sharps.

Proof.

- 1° Let $p \in \mathbb{P}_{\max}$.
- 2° By 3.2-I, for every $x \in \mathbb{R} \cap p$, there is $x^\sharp$.
- 3° Let $i : p \prec p^*$ be a generic ω_1 -iteration. Then $x^\sharp p = i(x^\sharp p) = x^\sharp p^*$.
- 4° Since $\omega_1^V \in p^*$, we have $x^\sharp = x^\sharp p^* = x^\sharp p \in p$.

□

III. **Lemma.** Suppose that:

- a. $W \models \text{ZFC}$;
- b. $\kappa := \text{Ord}^W < \omega_1^V$
- c. $M \in W$
- d. (M, I) is a pre-condition.

Then for every generic iteration M^* of (M, I) of length $< \kappa$, we have that $\text{Ord}^{M^*} < \kappa$. In particular, if $i : (M, I) \prec (M^*, I^*)$ is a κ -iteration, then $\omega_1^{M^*} = \kappa$.

Proof.

- 1° Let $\eta \in (|M|, \kappa)$ and $G \rightsquigarrow W^{\text{Col}(\omega, \eta)}$ be arbitrary.
- 2° Inside $W[G]$, the set of all Ord^{M^*} , where M^* is a generic η -iteration of (M, I) is Σ_1^I .
- 3° Thus, there is $\delta_\eta < \omega_1^{W[G]} < \kappa$ such that $\text{Ord}^{M^*} < \delta_\eta$ for every generic η -iteration M^* of (M, I) inside $W[G]$.
- 4° **Claim.** For every generic η -iteration M^* of (M, I) in V , we have $\text{Ord}^{M^*} < \delta_\eta$.

Proof.

- 1' Assume otherwise.
- 2' Then there is a generic η -iteration M^* of (M, I) in V such that $\text{Ord}^{M^*} \geq \delta_\eta$.
- 3' Let $x \in \mathbb{R}^{W[G]}$ code recursively pre-condition (M, I) and ordinals η, δ_η .
- 4' Statement $\ulcorner$ there is a generic η -iteration M^* of (M, I) such that $\text{Ord}^{M^*} \geq \delta_\eta \urcorner$ is a $\Sigma_1^1(x)$.
- 5' By absoluteness, this statement holds inside $W[G]$. This contradicts 3°.

□

- 5° The conclusion follows from the fact that $\delta_\eta < \kappa$.

□

IV. **Lemma.** Let $(p_i : i < \omega)$ be a decreasing sequence of $\mathbb{P}_{\max}$ -conditions and let $A := \cup\{a^{p_i} : i < \omega\}$. Then there exists a unique curious sequence $\mathcal{M}$ such that $\mathcal{M}_{i+1} \models \ulcorner (\mathcal{M}_i, A) \text{ is a generic iteration of } p_i \in H_{\omega_1} \urcorner$ and $\omega_1^{\mathcal{M}} = \sup\{\omega_1^{p_i} : i < \omega\}$.

Proof.

- 1° Let $j_{i,i+1} : p_i \prec p_i^{i+1}$ be the iteration of p_i given by p_{i+1} ; we have $j_{i,i+1} \in p_{i+1}$.
- 2° Let $j_{i,i+2} := j_{i+1,i+2}(j_{i,i+1}) : p_i \prec j_{i+1,i+2}(p_i^{i+1}) =: p_i^{i+2}$. This is the iteration of p_i given by p_{i+2} and $j_{i,i+2} \in p_{i+2}$. Furthermore, $j_{i,i+1}$ is an initial segment of $j_{i,i+2}$.
- 3° Continuing in this manner, we obtain iterations $j_{i,i+k} : p_i \prec p_i^{i+k}$ satisfying:
 - a. $j_{i,i+k}$ is the iteration of p_i given by p_{i+k} ;
 - b. $j_{i,i+k} \in p_{i+k}$
 - c. $j_{i,i+k}$ is an initial segment of $j_{i,i+l}$ for $k < l$.
- 4° Keeping 3°c in mind, we define $j_{i,\omega} : p_i \prec p_i^\omega$ as the appropriate direct limit.

5° It is now easily seen that

$$\mathcal{M} := ((M^{p_i^\omega}, I^{p_i^\omega}) : i < \omega)$$

is a required curious sequence.

6° The uniqueness follows from 1-XII.

□

V. **Definition.** The curious sequence from the previous lemma shall be referred to as *the curious sequence given by* $(p_i : i < \omega)$. The corresponding iterations are the unique generic iterations $j^i : p_i \prec (\mathcal{M}_i, A)$.

VI. **Lemma.** Let $(p_i : i < \omega)$ be a decreasing sequence of $\mathbb{P}_{\max}$ -conditions and let $\mathcal{M}$ be the curious sequence given by it. Then $\mathcal{M}$ is iterable.

Proof.

1° We will show that for every $i < \omega$, there is $x_i \in \mathbb{R} \cap \mathcal{M}_{i+1}$ such that

- a. $x_i^\# \in \mathcal{M}_{i+1}$
- b. the first x_i -indiscernible above $\omega_1^{\mathcal{M}}$ is $\geq \text{Ord}^{\mathcal{M}_i}$
- c. if $\mathcal{M}_i \models \ulcorner C \text{ is a club in } \omega_1 \urcorner$, then C contains all x_i -indiscernibles below $\omega_1^{\mathcal{M}}$ starting from some.

The conclusion then follows from 3.1-XV.

2° Let x_i be a real coding p_i .

3° Since $x_i \in p_{i+1}$, we have $x_i^\# \in p_{i+1}$ (by 3.2-II).

4° Consequently, $x_i, x_i^\# \in \mathcal{M}_{i+1}$, verifying 1°a.

5° Let ι be the $(\omega_1^{\mathcal{M}} + 1)^{\text{th}}$ indiscernible of $L[x_i]$.

6° $\mathcal{M}_i$ is a generic iteration of p_i of length $\omega_1^{\mathcal{M}} + 1 < \iota$.

7° By 3.2-III applied to model $L_\iota[x_i]$, we have $\text{Ord}^{\mathcal{M}_i} < \iota$, verifying 1°b.

8° Let now C be an $\mathcal{M}_i$ -club in $\omega_1^{\mathcal{M}}$. There is $\alpha < \omega_1^{\mathcal{M}}$ and a club C' in the α^{th} iterate of p_i such that C' is mapped to C by the iteration mapping.

9° Let $v > \alpha$ be an arbitrary x_i -indiscernible. By 3.2-III applied to model $L_v[x_i]$, v is the ω_1 of the v^{th} iterate of p_i .

10° Since C' is a club in the α^{th} iterate of p_i , we conclude that its image C contains the ω_1 of every iterate of p_i starting from the α^{th} one. In particular, $v \in C$, verifying 1°c.

□

VII. **Theorem.** Suppose that $x^\dagger$ exists for every $x \in \mathbb{R}$. Then $\mathbb{P}_{\max}$ is countably closed.

Proof.

- 1° Let $(p_i : i < \omega)$ be a decreasing sequence of $\mathbb{P}_{\max}$ conditions. We want to find $p \in \mathbb{P}_{\max}$ stronger than all p_i 's.
- 2° Let $\mathcal{M}$ be the curious sequence given by $(p_i : i < \omega)$, with $(j^i : i < \omega)$ the corresponding iterations. By 3.2-VI, $\mathcal{M}$ is iterable.
- 3° By 2-VII, there is a pre-condition (M, I) that contains the real coding $\mathcal{M}$ and such that $M \models \text{MA}_{\omega_1}$.
- 4° By 3.1-XVI inside M , there is an iteration $k : \mathcal{M} \rightarrow \mathcal{N}$ such that $I^{\mathcal{N}_i} = I \cap \mathcal{N}_i$ (for $i < \omega$).
- 5° Let $A := j^i(a^{p_i})$ (for some/any $i < \omega$). Let us set $a := k^i(A)$ (for some/any i). Then $p := (M, \in, I, a) \in \mathbb{P}_{\max}$.
- 6° $k^{i+1}(j^i) : p_i \prec k^{i+1}(\mathcal{M}_i) = \mathcal{N}_i$ is a generic iteration of p_i inside p of length ω_1^p .
- 7° We have

$$k^{i+1}(j^i)(a^{p_i}) = k^{i+1}(j^i(a^{p_i})) = k^{i+1}(A) = a,$$

$$k^{i+1}(j^i)(I^{p_i}) = k^{i+1}(j^i(I^{p_i})) = k^{i+1}(I^{\mathcal{M}_i}) = I^{\mathcal{N}_i} = I \cap \mathcal{N}_i.$$
- 8° We conclude $p < p_i$ for every $i < \omega$.

□

VIII. **Corollary.** Suppose that every real has the dagger. Then for $g \rightsquigarrow V^{\mathbb{P}_{\max}}$, we have $\omega_1^{V[g]} = \omega_1^V$, $\mathbb{R}^{V[g]} = \mathbb{R}^V$, and $\mathbb{P}_{\max}^{V[g]} = \mathbb{P}_{\max}^V$.

4 Subsets of ω_1 inside $L(\mathbb{R})^{\mathbb{P}_{\max}}$

4.1 Some Facts from Descriptive Set Theory

- I. **Notation.** $\mathbb{N}^{\text{III}}$ denotes the structure of the third-order arithmetic. The domain of first-order objects is ω , that of second-order objects is $\mathbb{R}$, while third-order objects are elements of $P(\mathbb{R})$.
- II. **Solovay's Basis Theorem.** Assume $V = L(\mathbb{R}) \models \text{ZF} + \text{DC}$. Let $\Phi(X)$ be a Σ_1^2 formula with X a third-order variable, and suppose that $\mathbb{N}^{\text{III}} \models \exists X \Phi(X)$. Then there is a Δ_1^2 subset $A \subseteq \mathbb{R}$ such that $\mathbb{N}^{\text{III}} \models \Phi[A]$.

Proof. Theorem 2.24 of [Jackson, 2010]. □

- III. **Definition.** Set $A \subseteq \mathbb{R}^k$ is *Suslin* if there exists $\gamma < \text{Ord}$ and a tree T on $\omega^k \times \gamma$ such that $A = p[T]$.
- IV. **Remark.** If $V \models \text{AC}$, then every set of reals is Suslin. This might not be the case if AC fails.
- V. **Theorem.** Assume $V = L(\mathbb{R}) \models \text{AD}$. Then every Σ_1^2 set $A \subseteq \mathbb{R}^k$ is Suslin.

Proof.

- 1° Let A be Σ_1^2 .
- 2° By Theorem 1 of [Martin and Steel, 1983], A has a scale.
- 3° By Lemma 2.5 of [Jackson, 2010], A is then Suslin.

□

- VI. **Proposition.** Assume $V \models \text{ZF} + \text{DC} + \text{AD}$. Then $\omega_1 \rightarrow (\omega_1)_{\mathbb{R}}^{\omega}$.

Proof. Exercise 28.13 of [Kanamori, 2008]. □

- VII. **Proposition.** Assume $V \models \text{ZF} + \text{DC} + \text{AD}$. Let $X \subseteq L$ be arbitrary. Then there is an inner model M such that $X \in M$ and such that there is $\kappa < \omega_1^V$ satisfying $M \models \ulcorner \kappa \text{ is measurable} \urcorner$.

Proof.

- 1° For an increasing $f : \omega \rightarrow \omega_1$, let

$$\kappa_f := \sup_{n < \omega} f(n), \quad F_f := \{C \subseteq \kappa_f : (\exists n_0 < \omega)(\forall n \geq n_0)(f(n) \in C)\}.$$

Thus, $\kappa_f < \omega_1$ and F_f is a filter on κ_f .

2° Let $M_f := L^{F_f}(X) = L^{F_f}[X]$ and let $U_f := F_f \cap M_f \in M_f$. We are done if we find f such that U_f is an ω_1 -complete ultrafilter in the sense of M_f , i.e. such that

$$\phi(f) : (\forall g \in \omega^{\kappa_f} \cap M_f)(\exists n < \omega)(g^{-1}(n) \in U_f).$$

3° We define a coloring $c : (\omega_1)^\omega \rightarrow \mathbb{R}$ by assigning to an increasing $f : \omega \rightarrow \omega_1$ a real $c(f)$ determined in the following manner.

- If $\phi(f)$, then $c(f) := (0 : n < \omega)$.
- If $\neg\phi(f)$, then let g_f be the minimal counter example, in the sense of the canonical order on $L^{F_f}[X]$. We set

$$c(f)(n) := \begin{cases} 1, & n = 0 \\ g_f(f(n-1)), & n > 0. \end{cases}$$

4° By 4.1-VI, there is an $H \in [\omega_1]^{\omega_1}$ homogeneous for c .

5° **Claim.** $c \upharpoonright (H)^\omega \equiv (0)_{n < \omega}$

Proof.

- 1' Assume otherwise. Let x be the value of c on $(H)^\omega$. We have $x(0) = 1$.
- 2' We will consider the cases where x is constant and where it is not constant, showing that both lead to a contradiction.
- 3' Suppose that x is constant and let $f \in (H)^\omega$ be arbitrary.
- 4' Since $c(f)(0) = x(0) = 1$, we have that $\neg\phi(f)$ and that g_f witnesses this fact.
- 5' Since $g_f(f(n)) = x(n+1)$ ($n < \omega$), we have that $g_f \circ f$ is constant.
- 6' This means that that g_f is constant on $\text{im } f \in F_f$. This contradicts 5°4'.
- 7' Suppose now that x is not constant, i.e. $x(n_0) \neq x(0)$ for some $n_0 > 0$. Let $f \in (H)^\omega$ be arbitrary.
- 8' Let $f' := (f(n_0 + n) : n < \omega) \in (H)^\omega$. We have

$$g_{f'}(f'(0)) = x(0) \neq x(n_0) = g_f(f(n_0)) = g_f(f'(0)),$$

implying $g_{f'} = g_f$.

- 9' On the other hand, clearly $F_f = F_{f'}$, so the models $L^{F_f}[X]$ and $L^{F_{f'}}[X]$ are the same, yielding $g_f = g_{f'}$. This is a contradiction. □

6° Thus, for every $f \in (H)^\omega$, we have $\phi(f)$. Since there is at least one such f , it can be chosen as to witness 2°. □

VIII. **Corollary.** Assume $V \models \text{ZF} + \text{DC} + \text{AD}$. Then every real has the dagger.

Proof.

- 1° Let $x \in \mathbb{R}$ be arbitrary. By 4.1-VII, there is an inner model $M \ni x$ and an M -measurable $\gamma < \omega_1^V$.
- 2° Since ω_1^V is measurable, there is an embedding $j : M \prec N$ with $\text{crit } j = \omega_1^V$.
- 3° This suffices for the conclusion that $x^\dagger$ exists.

□

IX. Proposition. Assume $V \models \text{ZF} + \text{DC} + \text{AD}$. Then the following holds.

- a. ω_1^V is inaccessible in every inner model.
- b. ω_1^V is a limit of inaccessibles in every inner model.

Proof.

Part IX.a.

- 1° Let M be an arbitrary inner model. Since ω_1^V is measurable, there is an embedding $j : M \prec N$ with $\text{crit } j = \omega_1^V$.
- 2° If $\gamma < \omega_1^V$ and $f = (\alpha_\xi : \xi < \gamma) \in (\omega_1^V)^\gamma \cap M$, then $j(f) = f$ is not cofinal in $j(\omega_1^V) > \omega_1^V$.
- 3° By elementarity, f is not cofinal in ω_1^V .
- 4° We conclude that ω_1^V is regular in M .
- 5° There is no injection $\omega_1^V \rightarrow \mathbb{R}$.
- 6° Thus, for every $\gamma < \omega_1^V$, there is no injection $\omega_1^V \rightarrow P^V(\gamma)$. Consequently, $M \models |P(\gamma)| < \omega_1^V$.
- 7° We conclude that ω_1^V is a strong limit in M .

Part IX.b.

- 8° Let M be an arbitrary inner model. Since ω_1^V is measurable, there is an embedding $j : M \prec N$ with $\text{crit } j = \omega_1^V$ and $\omega_1^V = j(\text{id}_{\omega_1^V})(\omega_1^V)$.
- 9° By IX.a, $N \models \ulcorner \omega_1^V \text{ is inaccessible} \urcorner$.
- 10° Thus, $\omega_1^V \in j(\{\alpha < \omega_1^V : M \models \ulcorner \alpha \text{ is inaccessible} \urcorner\})$ and consequently, $\{\alpha < \omega_1^V : M \models \ulcorner \alpha \text{ is inaccessible} \urcorner\}$ is unbounded in ω_1^V .

□

4.2 A Stronger Iterability Condition

- I. **Definition.** Let (M, I) be a pre-condition and let $A \subseteq \mathbb{R}$. Then (M, I) is *A-iterable* if $A \cap M \in M$ and for every iteration $j : (M, I) \prec (M^*, I^*)$ of countable length, we have $j(A \cap M) = A \cap M^*$.
- II. **Remark.** If (M, I) is *A-iterable*, then for every ω_1 -iteration $j : (M, I) \prec (M^*, I^*)$, we have $j(A \cap M) = A \cap M^*$.

III. **Proposition.** In $V \models \text{ZFC}$, there is $A \subseteq \mathbb{R}$ such that no pre-condition is A -iterable.

Proof.

- 1° There are stationary sets S, T such that $\omega_1 = S \cup T$.
- 2° Let A be the set of reals that code an ordinal in S . Let B be the set of reals that code an ordinal in T .
- 3° Suppose towards contradiction that there is an A -iterable pre-condition (M, I) . It is easily seen that (M, I) is also B -iterable.
- 4° Let S' be the set of ordinals coded by reals in $A \cap M \in M$; we have $S' \in M$. Let T' be the set of ordinals coded by reals in $B \cap M \in M$; we have $T' \in M$.
- 5° **Claim.** S' is I -positive in M .

Proof.

- 1' There is an iteration $j : (M, I) \prec (M^*, I^*)$ such that $I^* = \text{NS} \cap M^*$.
- 2' Since $A = j(A \cap M)$, we conclude $S = j(S') \in M$.
- 3' Thus, $M^* \models S \notin I^*$.
- 4' By elementarity, $M \models S' \notin I$.

□

- 6° Similarly, T' is I -positive in M .
- 7° There are now ω_1 -iterations $i : (M, I) \prec (M^*, I^*)$ and $j : (M, I) \prec (N^*, J^*)$ such that their first filters contain S' and T' , respectively.
- 8° This means that $\omega_1^M \in i(S') = S$, as well as $\omega_1^M \in j(T') = T$. This obviously a contradiction.

□

IV. We will next show that under the assumption $V = L(\mathbb{R}) \models \text{AD}$, there are A -iterable pre-conditions for every $A \subseteq \mathbb{R}$.

V. **Theorem.** Assume $V \models \text{ZFC}$. Let κ be regular satisfying $2^{<\kappa} = \kappa$. Then there is a c.c.c. poset $\mathbb{P} \subseteq \kappa$ such that $V^{\mathbb{P}} \models \text{MA} + 2^\omega = \kappa$.

Proof. This is essentially established in the proof of theorem 16.13 of [Jech, 2003].

□

VI. **Lemma.** Assume $V \models \text{ZFC}$. Suppose that κ is measurable and let $\mathbb{P} := \text{Col}(\kappa^+, 2^\kappa)$. Then inside $V^{\mathbb{P}}$, κ remains measurable and $2^\kappa = \kappa^+$.

Proof.

- 1° Since $\mathbb{P}$ is κ -closed, we have $P(\kappa)^V = P(\kappa)^{V^{\mathbb{P}}}$.

- 2° Thus, if μ is a normal ultrafilter on κ in the sense of V , it is an ultrafilter on κ in the sense of $V^{\mathbb{P}}$.
- 3° The fact that μ is normal in $V^{\mathbb{P}}$ readily follows from the fact that $V^{\kappa} \cap V^{\mathbb{P}} = V^{\kappa} \cap V$.
- 4° The cardinality conclusion is a standard forcing fact.

□

VII. Theorem. Assume $V = L(\mathbb{R}) \models \text{AD}$. Then for every $A \subseteq \mathbb{R}$, there exists A -iterable pre-condition (M, I) satisfying

$$(H(\omega_1)^M, \in, A \cap M) \prec (H(\omega_1), \in, A).$$

Furthermore, for any $x \in \mathbb{R}$, pre-condition (M, I) can be chosen so that $x \in M$.

Proof.

- 1° The second part clearly follows from the first part. Assume towards contradiction that the first part fails. Then there is an $A \subseteq \mathbb{R}$ such that there is no A -iterable pre-condition (M, I) with

$$(H(\omega_1)^M, \in, A \cap M) \prec (H(\omega_1), \in, A).$$

- 2° By 4.1-II, there is such A which is Δ_1^2 . (Recall that DC holds in the case $V = L(\mathbb{R}) \models \text{AD}$ by the result of [Kechris, 1984].)
- 3° Since $\{M \in [H_{\omega_1}]^{\omega} : (M, \in, A \cap M) \prec (H_{\omega_1}, \in, A)\}$ is club, there is $F' : H_{\omega_1}^{<\omega} \rightarrow H_{\omega_1}$ such that

$$(\forall M \in [H_{\omega_1}]^{\omega})(F'[M^{<\omega}] \subseteq M \implies (M, \in, A \cap M) \prec (H_{\omega_1}, \in, A)).$$

- 4° Since $\mathbb{R}$ codes H_{ω_1} , there is $F : \mathbb{R} \rightarrow \mathbb{R}$ such that for every transitive $M \in [H_{\omega_1}]^{\omega}$ satisfying ZFC^- , we have

$$F[\mathbb{R} \cap M] \subseteq \mathbb{R} \cap M \implies (M, \in, A \cap M) \prec (H_{\omega_1}, \in, A).$$

- 5° By 4.1-II, there is such an F which is Δ_1^2 .
- 6° By 4.1-V, sets $A, A^{\mathbb{G}}, F$ are Suslin. Thus, there are an ordinal γ , trees T, T' on $\omega \times \gamma$, and a tree U on $\omega \times \omega \times \gamma$ such that $A = p[T]$, $A^{\mathbb{G}} = p[T']$, and $F = p[U]$.
- 7° **Claim.** There are P, λ, I such that:
 - a. P is an inner model and $P \models \text{MA}_{\omega_1}$;
 - b. $\lambda < \omega_1^V$ is P -inaccessible;
 - c. $P \models \text{``}I \text{ is a precipitous ideal on } \omega_1 \text{''}$;
 - d. $T, T', U \in P$.

Proof.

- 1' By 4.1-VII, there is an inner model N and an ordinal $\kappa < \omega_1^V$ such that $T, T', U \in N$ and $N \models \ulcorner \kappa \text{ is measurable} \urcorner$.
- 2' By 4.1-IX, there is $\lambda \in (\kappa, \omega_1^V)$ such that $N \models \ulcorner \lambda \text{ is inaccessible} \urcorner$.
- 3' Let $\mathbb{P} := \text{Col}(\kappa^+, 2^\kappa)^N$.
- 4' $N^\mathbb{P} \models \ulcorner \kappa \text{ is measurable} \urcorner \wedge 2^\kappa = \kappa^+$ (cf. 4.2-VI).
- 5' Let $\mathbb{Q} := \text{Col}(\omega, < \kappa)$.
- 6' Inside $N^{\mathbb{P} * \mathbb{Q}}$, we have:
 - a. $\omega_1 = \kappa$
 - b. $2^{\omega_1} = \omega_2$
 - c. there is a precipitous ideal on ω_1 ;
 - d. λ is inaccessible.
- 7' By 4.2-V, there is $\dot{\mathbb{S}} \in N \restriction \lambda$ such that $N^{\mathbb{P} * \mathbb{Q}} \models \ulcorner \dot{\mathbb{S}} \text{ is c.c.c.} \urcorner$ and $N^{\mathbb{P} * \mathbb{Q} * \dot{\mathbb{S}}} \models \text{MA} + 2^\omega = \omega_2$.
- 8' Inside $N^{\mathbb{P} * \mathbb{Q} * \dot{\mathbb{S}}}$:
 - a. $2^\omega = \omega_2$
 - b. MA
 - c. there is a precipitous ideal on $\omega_1 = \kappa$;
 - d. λ is inaccessible.
 (For c, we apply 2-VI.)
- 9' There is $G \rightsquigarrow N^{\mathbb{P} * \mathbb{Q} * \dot{\mathbb{S}}}$. Let $P := N[G]$.

□

- 8° Let $M := P \restriction \lambda$. By 2-V, (M, I) is a pre-condition.
- 9° Since $U \in P$, we have $F[\mathbb{R} \cap M] \subseteq \mathbb{R} \cap M$. Thus,

$$(H(\omega_1)^M, \in, A \cap M) \prec (H(\omega_1), \in, A).$$

It remains to verify that (M, I) is A -iterable.

- 10° Since $T \in P$, we have $A \cap M = p[T]^P \in M$.
- 11° Let $j : (M, I) \prec (M^*, I^*)$ be a generic iteration of countable length and let us verify $j(A \cap M) = A \cap M^*$. Let $k : (P, I) \prec (P^*, I^*)$ be the iteration of (P, I) obtained using the same generics as for j . (The well-foundedness follows from 2-IV.)
- 12° The following are readily verifiable:
 - a. $p[T] \cap P \subseteq p[j(T)] \cap P^*$
 - b. $p[T'] \cap P \subseteq p[j(T')] \cap P^*$
 - c. $(p[T] \cap P) \cup (p[T'] \cap P) = \mathbb{R} \cap P$
 - d. $(p[j(T)] \cap P^*) \cap (p[j(T')] \cap P^*) = \emptyset$
 - e. $\mathbb{R} \cap P = \mathbb{R} \cap P^*$.
 We conclude $p[j(T)] \cap P^* = p[T] \cap P$.
- 13° Thus, $j(A \cap M) = j(p[T] \cap M) = p[j(T)] \cap M^* = p[j(T)] \cap P^* = p[T] \cap P = A \cap P = A \cap P^*$, confirming A -iterability.

□

4.3 The Generic Power Set of ω_1

I. **Definition.** For $g \rightsquigarrow V^{\mathbb{P}_{\max}}$, we define

$$A_g := \bigcup_{p \in g} a^p \subseteq \omega_1^V.$$

II. **Proposition.** Assume that every real has the dagger. Let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$ and work in $V[g]$. Then:

- a. A_g is unbounded in ω_1 ;
- b. for every $p \in g$, $A_g \cap \omega_1^p = a^p$;
- c. for every $p \in g$, there is a unique generic ω_1 -iteration $i : p \prec p^*$ such that $i(a^p) = A_g$; every iterate of p in this iteration belongs to g .

Proof.

Part a.

- 1° Let $\alpha < \omega_1$ be arbitrary.
- 2° Let $x \in \mathbb{R}$ code a well-ordering on ω of order type α . Set $\{p \in \mathbb{P}_{\max} : x \in p\}$ is dense.
- 3° Thus, there is $p \in g$ with $x \in p$.
- 4° This means $\alpha < \omega_1^p$.
- 5° By 1-X, there is $\beta \in a^p \subseteq A_g$ such that $\alpha < \beta$.

Part b.

- 6° It suffices to show that for $p, q \in g$ with $\omega_1^p \leq \omega_1^q$, we have $a^q \cap \omega_1^p = a^p$.
- 7° Let $r \in g$ with $r < p, q$.
- 8° Let $i : p \prec r$ and $j : q \prec r$ be the iterations of p, q given by r .
- 9° $a^p = i(a^p) \cap \omega_1^p = a^r \cap \omega_1^p = j(a^q) \cap \omega_1^p = j(a^q) \cap \omega_1^q \cap \omega_1^p = a^q \cap \omega_1^p$

Part c.

- 10° By 1-XII, there is at most one such iteration. Let us show that there is at least one. (It will be clear that every iterate is in g .)
- 11° We define by recursion generic iterations $i_{\alpha, \beta} : p_\alpha \prec p_\beta$ ($\alpha \leq \beta < \omega_1$) such that:
 - a. $p_0 := p$
 - b. $p_\alpha \in g$ (all α)
 - c. $i_{\alpha, \beta}$ is of countable length, which is at least 1, for all $\alpha < \beta$;
 - d. family $(i_{\alpha, \beta})_{\alpha \leq \beta}$ commutes.
- 12° Let us consider recursively the case $\alpha + 1$. We want to define $i_{\alpha, \alpha+1} : p_\alpha \prec p_{\alpha+1}$.
- 13° Let $q \in g$ be such that $\omega_1^q > \omega_1^{p_\alpha}$. Let $p_{\alpha+1} \in g$ be such that $p_{\alpha+1} < p_\alpha, q$.

- 14° Let $i_{\alpha, \alpha+1} : p_\alpha \prec p_{\alpha+1}$ witness $p_{\alpha+1} < p_\alpha$.
- 15° This was the successor step. At the limit step, we take the direct limit. The recursion is now complete.
- 16° The direct limit of $(p_\alpha, i_{\alpha, \beta} : \alpha \leq \beta < \omega_1)$ provides a generic iteration $i : p \prec p^*$ of length ω_1 .
- 17° **Claim.** i is as required, i.e. $i(a^p) = A_g$.

Proof.

- 1' For every $\alpha < \omega_1$, we have

$$i(a^p) \cap \omega_1^{p_\alpha} = i_{0, \alpha}(a^p) = a^{p_\alpha} = A_g \cap \omega_1^{p_\alpha}.$$

The last equality follows from the fact that $p_\alpha \in g$.

- 2' This suffices, for set $\{\omega_1^{p_\alpha} : \alpha < \omega_1\}$ is cofinal in ω_1 .

□

□

- III. **Definition.** For $g \rightsquigarrow V^{\mathbb{P}_{\max}}$ and $p \in g$, the unique generic ω_1 -iteration $i : p \prec p^*$ such that $i(a^p) = A_g$ is called *the iteration of p given by g* .
- IV. **Remark.** Suppose that $q < p$ and let $i_{p, q} : p \prec p^*$ be the iteration of p given by q . Let g be $\mathbb{P}_{\max}$ -generic and let $i_{q, g} : q \prec q^*$ be the iteration of q given by g . Then $i_{q, g}(i_{p, q}) : p \prec i_{q, g}(p^*)$ is the iteration of p given by g . In other words,

$$q \Vdash \ulcorner i_{p, q} \urcorner \text{ is an initial segment of } i_{p, g} \urcorner.$$

- V. **Definition.** Let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$ and let $i_{p, g} : p \prec p_g$ denote the iteration of $p \in g$ given by g . We define

$$P(\omega_1)_g := \bigcup_{p \in g} P(\omega_1)^{p_g}.$$

- VI. **Lemma.** Assume that $x^\dagger$ exists for every $x \in \mathbb{R}$. Let J be a normal ideal on ω_1 , let $p \in \mathbb{P}_{\max}$, and let $(D_\alpha : \alpha < \omega_1)$ be dense subsets of $\mathbb{P}_{\max}$. Then there exist an iteration $i : p \prec p^*$ and a sequence $(p_\alpha : \alpha < \omega_1)$ of conditions such that:

- a. $p_0 = p$ and $p_\beta < p_\alpha$ for $\alpha < \beta$;
- b. the iteration of p given by p_α is an initial segment of i ;
- c. $p_{\alpha+1} \in D_\alpha$ for $\alpha < \omega_1$
- d. $i(I^p) = J \cap p^*$.

Proof.

- 1° This is a variation on the theme of 2-X. We will define sequence $(p_\alpha : \alpha < \omega_1)$ by recursion on α . We also fix enumerations $\{B_i^\alpha : i < \omega\} = P(\omega_1)^{p_\alpha} - I^{p_\alpha}$ as we go.
- 2° Embedding $i_{\alpha,\beta} : p_\alpha \prec p_\beta$ will be the iteration of p_α given by p_β (for $\alpha \leq \beta$).
- 3° Let $\{A_{i,\alpha} : i < \omega, \alpha < \omega_1\}$ be pairwise disjoint J -positive sets whose union is ω_1 .
- 4° At stage $\alpha = 0$, let $p_0 := p$.
- 5° At stage $\alpha + 1$, we choose arbitrary $p_{\alpha+1} < p_\alpha$ with $p_{\alpha+1} \in D_\alpha$.
- 6° Let us now consider limit stage γ . Let $h_\gamma : \omega \rightarrow \gamma$ be an increasing cofinal function and let $\mathcal{M}_\gamma$ be the curious sequence given by $(p_{h(i)} : i < \omega)$, with $j_\gamma^i : p_{h(i)} \prec (\mathcal{M}_\gamma)_i$ the corresponding iterations.
- 7° By 2-VII, there is a pre-condition (M, I) containing a real coding $\mathcal{M}_\gamma$ such that $M \models \text{MA}_{\omega_1}$.
- 8° $\omega_1^{\mathcal{M}_\gamma}$ belongs to exactly one $A_{s,\alpha}$. Let us choose an $\mathcal{M}_\gamma$ -generic G_γ satisfying: if $t < \omega$ is such that $h_\gamma(t) > \alpha$, then $j_\gamma^t(i_{\alpha,h_\gamma(t)}(B_s^\alpha)) \in G_\gamma$. (Note that here we are talking here about only one set when $\alpha < \gamma$, and we are not requiring anything otherwise.) Let $\mathcal{M}' := \text{Ult}(\mathcal{M}_\gamma, G_\gamma)$.
- 9° By 3.2-VI, $\mathcal{M}_\gamma$ is iterable. Consequently, $\mathcal{M}'$ is also iterable.
- 10° By 3.1-XVI inside M , there is an ω_1 -iteration $k : \mathcal{M}' \prec \mathcal{N}$ such that $I^{\mathcal{N}_i} = I \cap \mathcal{N}_i$ for every $i < \omega$.
- 11° Let $a_\gamma := \cup\{a^{p_\alpha} : \alpha < \gamma\}$ and let $p_\gamma := (M, \in, I, k^0(j_\gamma^0(a_\gamma)))$. Clearly, $p_\gamma \in \mathbb{P}_{\max}$ and $p_\gamma < p_\alpha$ for every $\alpha < \gamma$.
- 12° This concludes the recursion. Iterations i_{α_0} are determined by commutative systems

$$(i_{\alpha,\beta} \upharpoonright p_{\alpha_0}^\alpha : \alpha_0 \leq \alpha \leq \beta < \omega_1)$$

of iterations. We set $i := i_0$.

- 13° **Claim.** $I^{p^*} = J \cap p^*$

Proof.

- 1' By 2-IX, we have $I^{p^*} \subseteq \text{NS}_{\omega_1} \subseteq J$.
- 2' Let us verify $(\supseteq)$. Let $X \in P(\omega_1)^{p^*} - I^{p^*}$ be arbitrary and let us show $X \in P(\omega_1) - J$.
- 3' We have that $X = i_\alpha(B_s^\alpha)$ for some $\alpha < \omega_1$ and some $s < \omega$.
- 4' $C := \{\omega_1^{\mathcal{M}_\gamma} : \gamma = \cup \gamma \in (\alpha, \omega_1)\}$ is club and consequently of J -measure one.
- 5' Since $A_{s,\alpha}$ is J -positive, set $S = C \cap A_{s,\alpha}$ is also J -positive.
- 6' Let us consider some $\omega_1^{\mathcal{M}_\gamma} \in S$. We have $\omega_1^{\mathcal{M}_\gamma} \in A_{s,\alpha}$ and $\alpha < \gamma$.
- 7' Thus, for sufficiently large t , we have $j_\gamma^t(i_{\alpha,h_\gamma(t)}(B_s^\alpha)) \in G_\gamma$ and consequently $\omega_1^{\mathcal{M}_\gamma} \in i_{\alpha,\gamma}(B_s^\alpha) \subseteq X$.

8' This means $S \subseteq X$.

9' Since S is J -positive, we have $X \in P(\omega_1)^{p^*} - J$.

□

□

VII. **Lemma.** Assume $g \rightsquigarrow V^{\mathbb{P}_{\max}}$, let $p \in g$, and let $i : p \prec p^*$ be the iteration of p given by g . Let $P \in p$ be a subset of $\mathbb{P}_{\max}^V$ and suppose that $(\forall q \in P)(p < q)$. Then $i(P) \subseteq g$.

Proof.

- 1° We denote iterates of i by p_ξ and embeddings by $i_{\xi, \eta}$.
- 2° Let $q \in i(P)$ be arbitrary. Then there is $\xi < \omega_1$ and $q' \in i_{0, \xi}(P)$ such that $q = i_{\xi, \omega_1}(q')$.
- 3° Since i is the iteration of p given by g , we conclude $p_\xi \in g$.
- 4° Since “ $q < p$ ” is first-order expressible in p and $p \models (\forall q \in P)(“q < p”)$, we conclude $p_\xi \models (\forall q \in i_{0, \xi}(P))(“q < p_\xi”)$.
- 5° Thus, $q' < p_\xi \in g$ and consequently $q' \in g$.
- 6° Since $(\forall q \in P)(p < q)$, we have $(\forall q \in P)(q \in H_{\omega_1}^p)$ and consequently $(\forall q \in i_{0, \xi}(P))(q \in H_{\omega_1}^{p_\xi})$.
- 7° Thus, $q' \in H_{\omega_1}^{p_\xi}$, which yields $q = i_{0, \xi}(q') = q' \in g$.

□

VIII. **Theorem.** Assume $V = L(\mathbb{R}) \models \text{AD}$ and $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Then $P(\omega_1)^{V[g]} = P(\omega_1)_g$.

Proof.

- 1° Let $\tau \in V^{\mathbb{P}_{\max}}$ be such that $\Vdash \tau \subseteq \omega_1$. We want to show $\Vdash \tau \in P(\omega_1)_G$.
- 2° Let $p \in \mathbb{P}_{\max}$ be arbitrary. We want to find some $p' < p$ such that $p' \Vdash \tau \in p'_G$ (cf. 4.3-V for the notation).
- 3° Let us fix standard recursive (partial) coding $f : \mathbb{R} \rightarrow H_{\omega_1}$. Let A denote set
$$\{x \in \mathbb{R} : (\exists p \in \mathbb{P}_{\max})(\exists \alpha < \omega_1^p)(\exists i < 2)(f(x) \downarrow = (p, \alpha, i) \wedge p \Vdash \chi_\tau(\alpha) = i)\}.$$
- 4° By 4.2-VII, there is an A -iterable pre-condition (M, I) containing a real coding p and satisfying

$$(H_{\omega_1}^M, \in, A \cap M) \prec (H_{\omega_1}, \in, A). \quad (1)$$

- 5° By (1), $M \models (\forall x \in \mathbb{R}) \exists y (y = x^\dagger)$. By 4.3-VI, inside M , there is a descending sequence $(p_\alpha : \alpha < \omega_1)$ of $\mathbb{P}_{\max}$ conditions, together with an iteration $i : p \prec p^*$, such that $p_0 = p$ and:

- a. $(\forall \alpha < \omega_1)(\alpha < \omega_1^{p_{\alpha+1}} \wedge p_{\alpha+1} \parallel \chi_\tau(\alpha))$
 - b. the iteration of p given by p_α is an initial segment of i ;
 - c. $i(I^p) = I \cap p^*$.
- 6° By elementarity, $(p_\alpha : \alpha < \omega_1^M)$ is a descending sequence of $\mathbb{P}_{\max}$ conditions (in V), mapping $i : p \prec p^*$ is an iteration, and the following holds:
- a. $p_0 = p$
 - b. $(\forall \alpha < \omega_1^M)(p_{\alpha+1} \parallel \chi_\tau(\alpha))$
 - c. the iteration of p given by p_α is an initial segment of i ;
 - d. $i(I^p) = I \cap p^*$.
- 7° Let $p' := (M, \in, I, i(a^p))$. We have $p' < p_\alpha$ for every $\alpha < \omega_1^M$ and in particular, $p' < p$.
- 8° By (1), we have

$$e := \{\alpha < \omega_1^M : p_{\alpha+1} \Vdash \alpha \in \tau\} \in P(\omega_1)^{p'}.$$

$$9^\circ e = \{\alpha < \omega_1 : (\exists x \in A \cap M)(f(x) \downarrow = (p_{\alpha+1}, \alpha, 1))\}$$

$$10^\circ \textbf{Claim. } p' \Vdash i_{p',g}(e) = \tau$$

Proof.

- 1' Let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$ with $g \ni p'$ be arbitrary. We want to verify $i_{p',g}(e) = \tau^g$.
- 2' Let $(q_\alpha : \alpha < \omega_1) = i_{p',g}((p_\alpha : \alpha < \omega_1^M))$.
- 3' We have

$$q_{\alpha+1} \Vdash \alpha \in \tau \iff (\exists x \in A)(f(x) \downarrow = (q_{\alpha+1}, \alpha, 1)) \quad (2)$$

$$\iff (\exists x \in A \cap p'_g)(f(x) \downarrow = (q_{\alpha+1}, \alpha, 1)) \quad (3)$$

$$\iff (\exists x \in i_{p',g}(A \cap M))(f(x) \downarrow = (q_{\alpha+1}, \alpha, 1)) \quad (4)$$

$$\iff \alpha \in i_{p',g}(e). \quad (5)$$

Explanations are as follows:

(2) follows by definition of A ;

(3) follows from the fact that $(q_{\alpha+1}, \alpha, 1) \in p'_g$;

(4) follows by A -iterability of p and the fact that p' is an iterate of p ;

(5) follows by 9°, 10°2', and elementarity of $i_{p',g}$.

4' Analogously, we establish

$$q_{\alpha+1} \Vdash \alpha \notin \tau \iff \alpha \notin i_{p',g}(e).$$

5' By 4.3-VII, we have $\{q_\alpha : \alpha < \omega_1\} \subseteq g$. Points 10°3' and 10°4' now yield $i_{p',g}(e) = \tau^g$.

□

11° Thus, p' is as required in 2°.

□

IX. **Lemma.** Let $p \in \mathbb{P}_{\max}$ and $A \subseteq \omega_1$. If the generic iteration $i : p \prec p^*$ satisfying

$$i(\omega_1^p) = \omega_1, \quad i(a^p) = A$$

exists, then it belongs to $L(p, A)$.

Proof.

- 1° Let $(g_\alpha : \alpha < \omega_1)$ be such iteration. By induction on α , we show that $g_\alpha \in L(p, A)$.
- 2° Assume inductively that $(g_\beta : \beta < \alpha) \in L(p, A)$ and let us show that $g_\alpha \in L(p, A)$.
- 3° We define f such that $L[a^p]^p \models f : \omega_1 \rightarrow P(\omega)$ by recursion.
 - Let $\text{code} : \omega \times \omega \rightarrow \omega$ be the standard bijection. For $n < \omega$, we set $f(n) := \{\text{code}(n, i) : i < \omega\}$.
 - For $\alpha \in [\omega, \omega_1^p)$, assume recursively that $f \upharpoonright \alpha$ is defined and almost disjoint. Then $f[\alpha]$ is not maximal. Let $f(\alpha)$ be the $<_{L[a^p]^p}$ -minimal subset of ω almost disjoint from every element of $f[\alpha]$.
- 4° By elementarity, inside $L[A]^{p^*}$, we have:
 - a. $i(f) : \omega_1 \rightarrow P(\omega)$
 - b. for $n < \omega$, we set $i(f) := \{\text{code}(n, i) : i < \omega\}$;
 - c. $i(f)$ is the $<_{L[A]}$ -minimal subset of ω almost disjoint from every element of $i(f)[\alpha]$.
- 5° Since $\omega_1^{L[A]^{p^*}} = \omega_1^{L[A]}$, we have inside $L[A]$:
 - a. $i(f) : \omega_1 \rightarrow P(\omega)$
 - b. for $n < \omega$, we set $i(f) := \{\text{code}(n, i) : i < \omega\}$;
 - c. $i(f)$ is the $<_{L[A]}$ -minimal subset of ω almost disjoint from every element of $i(f)[\alpha]$.
- 6° Let $i_{0,\alpha}$ be the iteration from p to p_α and i_{α,ω_1} the iteration from p_α to p^* along i (we have $i = i_{\alpha,\omega_1} \circ i_{0,\alpha}$).
- 7° Since $p_\alpha \models \text{MA}_{\omega_1}$, for every $S \in P(\omega_1)^{p_\alpha}$, there is $x_S \in P(\omega)^{p_\alpha}$ such that

$$(\forall \xi < \omega_1^{p_\alpha})(\xi \in S \iff |i_{0,\alpha}(f)(\xi) \cap x_S| = \omega).$$

- 8° By elementarity, for every $S \in P(\omega_1)^{p_\alpha}$,

$$(\forall \xi < \omega_1)(\xi \in i_{\alpha,\omega_1}(S) \iff |i(f)(\xi) \cap x_S| = \omega).$$

- 9° For every $S \in P(\omega_1)^{p_\alpha}$:

$$S \in g_\alpha \iff \omega_1^{p_\alpha} \in i_{\alpha,\omega_1}(S) \iff |i(f)(\omega_1^{p_\alpha}) \cap x_S| = \omega.$$

10° $i(f) \in L[A]$ since it is definable there; $x_S \in p_\alpha \in L(p, A)$ by the IH. Thus, the previous characterization implies $g_\alpha \in L(p, A)$.

□

X. **Theorem.** Assume $V = L(\mathbb{R}) \models \text{AD}$ and $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Then $V[g] = L(\mathbb{R}, A_g)$.

Proof.

1° $V[g] = L(\mathbb{R})[g] = L(\mathbb{R}, g)$

2° Thus, $V[g] = L(P(\omega_1)^{V[g]})$.

3° By 4.3-VIII, $P(\omega_1)^{V[g]} = \{i(x) : p \in \mathbb{P}_{\max}, x \in P(\omega_1)^p, i : p \prec p^* \text{ is an } \omega_1\text{-iteration satisfying } i(a^p) = A_g\}$.

4° 4.3-IX now implies $P(\omega_1)^{V[g]} \subseteq L(\mathbb{R}, A_g)$. Consequently, $V[g] = L(\mathbb{R}, A_g)$.

□

5 Properties of $L(\mathbb{R})^{\mathbb{P}_{\max}}$

- I. **Definition.** ψ_{AC} is the following statement: for every stationary co-stationary sets $A, B \subseteq \omega_1$, there exists a bijection $f : \omega_1 \rightarrow \gamma$ such that

$$\{\xi < \omega_1 : \xi \in A \iff \text{ot}(f[\xi]) \in B\}$$

contains a club.

- II. **Lemma.** Assume $V \models \text{ZFC}$. Let $x \in \mathbb{R}$ and assume $x^\sharp$ exists. Let γ be the $(\omega_1 + 1)^{\text{th}}$ indiscernible for $L[x]$ and let $f : \omega_1 \rightarrow \gamma$ be an arbitrary bijection. Then set $\{\alpha < \omega_1 : \text{ot}(f[\alpha]) \text{ is the least } x\text{-indiscernible above } \alpha\}$ contains a club.

Proof.

- 1° Let $h : \omega_1 \rightarrow \omega_1$ be defined as to map α to the least x -indiscernible above α .
 2° It suffices to verify the following: for arbitrary $G \rightsquigarrow V^{\mathbb{P}_{\text{NS}}}$, we have

$$[(\text{ot}(f[\alpha]) : \alpha < \omega_1)]_G = [h].$$

- 3° Let $j := j_G : V \prec \text{Ult}(V, G) = \mathfrak{M}$.
 4° Inside $\mathfrak{M}$, $[h] = j(h)(\omega_1^V)$ is the $(\omega_1^V + 1)^{\text{th}}$ indiscernible for $L[x]$. Since ω_1^V is well-founded, we conclude that $[h] = \gamma$.
 5° $[(\text{ot}(f[\alpha]) : \alpha < \omega_1)] = \text{ot}^{\mathfrak{M}}(j(f)[\omega_1^V])$
 6° $j(f)[\omega_1^V] = \{j(f)(\alpha) : \alpha < \omega_1^V\} = \{j(f(\alpha)) : \alpha < \omega_1^V\} = j[f[\omega_1^V]] = j[\gamma]$
 7° Thus, $[(\text{ot}(f[\alpha]) : \alpha < \omega_1)] = \text{ot}^{\mathfrak{M}}(j[\gamma]) = \gamma = [h]$.

□

- III. **Lemma.** Assume $V \models \text{ZFC}$ and let J be a normal ideal on ω_1 . Let (M, I) be a pre-condition and let $A, B \in P(\omega_1)^M - I$ with $A^{\mathbb{C}}, B^{\mathbb{C}} \in P(\omega_1)^M - I$. Then there is an ω_1 -iteration $j : (M, I) \prec (M^*, I^*)$ with $I^* = J \cap M^*$ and a bijection $f : \omega_1 \rightarrow \gamma$ such that

$$\{\xi < \omega_1 : \xi \in j(A) \iff \text{ot}(f[\xi]) \in j(B)\}$$

contains a club.

Proof.

- 1° This is a variation on the theme of [2-X](#). Let x be a real coding (M, I) . Let ι_α be the α^{th} indiscernible for $L[x]$. Let $(A_{i,\alpha} : i < \omega, \alpha < \omega_1)$ be pairwise disjoint J -positive sets whose union is ω_1 .

- 2° We define a generic iteration $(g_\alpha : \alpha < \omega_1)$ of (M, I) by recursion on α ; iterates are denoted by (M_ξ, I_ξ) , while embeddings are denoted by $i_{\xi, \eta}$. We fix enumerations $\{B_j^\alpha : j < \omega\} = P(\omega_1)^{M_\alpha} - I_\alpha$ as we go.
- 3° Suppose recursively that $(g_\xi : \xi < \gamma)$ have been defined. In particular, iterate (M_γ, I_γ) is defined.
- 4° If γ is not of the form ι_η or $\iota_{\eta+1}$ for a limit η , then generic g_γ is chosen with no restrictions.
- 5° If $\gamma = \iota_{\eta+1}$ for a limit η , then generic g_γ is chosen such that $i_{0, \gamma}(B) \in g_\gamma$ if and only if $i_{0, \iota_\eta}(A) \in g_{\iota_\eta}$.
- 6° It remains to consider case $\gamma = \iota_\eta$, where η is limit. Ordinal $\omega_1^{M_\gamma}$ belongs to exactly one $A_{j, \xi}$.
- 7° Choose $g_\gamma \subseteq P(\omega_1)^{M_\gamma} - I_\gamma$ which is M_γ -generic and satisfies: if $\xi \leq \gamma$, then $i_{\xi, \gamma}(B_j^\xi) \in g_\gamma$.
- 8° This concludes the recursion.
- 9° **Claim.** $I_{\omega_1} = J \cap M_{\omega_1}$

Proof.

- 1' By 2-IX, we have $I_{\omega_1} \subseteq \text{NS}_{\omega_1} \subseteq J$.
- 2' Let us verify $(\supseteq)$. Let $X \in P(\omega_1)^{M_{\omega_1}} - I_{\omega_1}$ be arbitrary and let us verify that $X \in P(\omega_1) - J$.
- 3' We have $X = i_{\xi, \omega_1}(B_j^\xi)$ for some $\xi < \omega_1$ and some $j < \omega$.
- 4' $C := \{\omega_1^{M_{\iota_\eta}} : \eta = \cup \eta \in [\xi, \omega_1)\}$ is club and consequently of J -measure one.
- 5' Since $A_{j, \xi}$ is J -positive, set $S := C \cap A_{j, \xi}$ is J -positive.
- 6' For $\omega_1^{M_{\iota_\eta}} \in S$, we have $\omega_1^{M_{\iota_\eta}} \in A_{j, \xi}$ and $\xi \leq \iota_\eta$. Thus, $i_{\xi, \iota_\eta}(B_j^\xi) \in g_{\iota_\eta}$ and consequently, $\omega_1^{M_{\iota_\eta}} \in i_{\xi, \iota_\eta+1}(B_j^\xi) \subseteq X$.
- 7' We conclude $S \subseteq X$. Since S is J -positive, we have $X \in P(\omega_1)^{M_{\omega_1}} - J$.

□

- 10° Let $j := i_{0, \omega_1}$, let $\gamma := \iota_{\omega_1+1}$, and let $f : \omega_1 \rightarrow \gamma$ be an arbitrary bijection.
- 11° **Claim.** $\{\xi < \omega_1 : \xi \in j(A) \iff \text{ot}(f[\xi]) \in j(B)\}$ contains a club.

Proof.

- 1' Let $C := \{\iota_{\omega_\alpha} : \alpha < \omega_1, \text{ot}(f[\iota_{\omega_\alpha}]) = \iota_{\omega_{\alpha+1}}\}$. By 5-III, this set contains a club.
- 2' By 3.2-III, we have $\omega_1^{M_\gamma} = \gamma$ for every $\gamma = \iota_\alpha$ ($\alpha < \omega_1$).

3' For $\gamma = \iota_{\omega_\alpha} \in C$, we have

$$\gamma \in j(A) \iff \omega_1^{M_\gamma} \in j(A) \quad (6)$$

$$\iff i_{0,\gamma}(A) \in g_\gamma \quad (7)$$

$$\iff i_{0,\iota_{\omega_\alpha+1}}(B) \in g_{\iota_{\omega_\alpha+1}} \quad (8)$$

$$\iff \omega_1^{M_{\iota_{\omega_\alpha+1}}} \in j(B) \quad (9)$$

$$\iff \iota_{\omega_\alpha+1} \in j(B) \quad (10)$$

$$\iff \text{ot}(f[\gamma]) \in j(B). \quad (11)$$

□

□

IV. **Theorem.** Assume $V = L(\mathbb{R}) \models \text{AD}$ and $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Then $V[g] \models \psi_{\text{AC}}$.

Proof.

1° We work in $V[g]$. Let $A, B \subseteq \omega_1$ be arbitrary stationary co-stationary sets.

2° By 4.3-VIII, there is $p \in g$ such that its iteration $i : p \prec p^*$ given by g satisfies $A, B \in p^*$.

3° We may assume w.l.o.g. that there are $A', B' \in p$ such that $A = i(A')$ and $B = i(B')$.

4° Let $D \subseteq \mathbb{P}_{\max}$ consist of all $q < p$ such for the iteration j of p given by q , we have the following inside q : there exists a bijection $f : \omega_1 \rightarrow \gamma$ such that

$$\{\xi < \omega_1 : \xi \in j(A) \iff \text{ot}(f[\xi]) \in j(B)\}$$

contains a club.

5° By 5-III, we have that D is dense. Indeed, for any $q_0 \leq p$, one can choose a pre-condition (N, J) with $q_0 \in H_{\omega_1}^N$ and apply 5-III within N to q_0 and to the appropriate images of A', B' .

6° There is $q \in g \cap D$. The conclusion now follows by elementarity of $i_{q,g}$.

□

V. **Lemma.** Assume $V = L(P(\omega_1)) \models \text{ZF} + \psi_{\text{AC}}$ and suppose that there is a partition of ω_1 into ω_1 disjoint stationary sets. Then $V \models \text{AC}$.

Proof.

1° It suffices to show that there is an injection $\Phi : P(\omega_1) - \{\emptyset, \omega_1\} \rightarrow \omega_2$.

2° Let $(S_\alpha : \alpha < \omega_1)$ be a partition of ω_1 into stationary sets and let $S := S_0$.

3° For non-empty $X \subset \omega_1$, we set

$$A_X := \bigcup_{\alpha \in X} S_\alpha$$

(which is stationary and co-stationary) and we set $\Phi(X)$ to be the minimal $\gamma < \omega_2$ such that there exist a bijection $f : \omega_1 \rightarrow \gamma$ and a club $C \subseteq \omega_1$ satisfying

$$(\forall \alpha \in C)(\alpha \in A_X \iff \text{ot}(f[\alpha]) \in S)$$

(which is correctly defined by ψ_{AC}).

4° Suppose that $\Phi(X_0) = \Phi(X_1)$ and let us show that $X_0 = X_1$.

5° Let $f_i : \omega_1 \rightarrow \Phi(X_i)$ be a bijection and let C_i be a club such that

$$(\forall \alpha \in C_i)(\alpha \in A_{X_i} \iff \text{ot}(f_i[\alpha]) \in S)$$

($i = 0, 1$).

6° Set $D := \{\alpha < \omega_1 : \text{ot}(f_0[\alpha]) = \text{ot}(f_1[\alpha])\}$ is club. Consequently, set $C_0 \cap C_1 \cap D$ is club.

7° For $\alpha \in (A_{X_0} \triangle A_{X_1}) \cap (C_0 \cap C_1 \cap D)$, we would have

$$\alpha \in A_{X_0} \iff \text{ot}(f_0[\alpha]) \in S \quad (12)$$

$$\iff \text{ot}(f_1[\alpha]) \in S \quad (13)$$

$$\iff \alpha \in A_{X_1}, \quad (14)$$

which is obviously impossible. Thus, $A_{X_0} \triangle A_{X_1}$ is non-stationary.

8° This implies that $X_0 \triangle X_1 = \emptyset$, which means $X_0 = X_1$.

□

VI. Proposition. Assume that every real has the dagger and let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Suppose that $p \in g$ and let $i_{p,g} : p \prec p^*$ be the iteration of p given by g . Then $I^{p^*} = \text{NS}_{\omega_1}^{V[g]} \cap p^*$.

Proof.

1° By 2-IX, we have $I^{p^*} \subseteq \text{NS}_{\omega_1}^{V[g]}$.

2° Let us verify ($\supseteq$). Let $S \in P(\omega_1)^{p^*} - I^{p^*}$ be arbitrary and let us show that S is stationary.

3° Let $C \subseteq \omega_1$ be a club and let us show that $C \cap S \neq \emptyset$.

4° The iterates in $i_{p,g}$ are denoted by p_ξ , while embeddings are denoted by $i_{\xi,\eta}$. There are $\alpha < \omega_1$ and $S' \in I^{p_\alpha}$ such that $i_{\alpha,\omega_1}(S') = S$.

5° Let $\dot{C}$ be a name for C and let $q \in g$ be such that $q < p, p_\alpha$ and $q \Vdash \ulcorner \dot{C} \text{ is club} \urcorner$.

6° **Claim.** $D := \{r \leq q : r \Vdash \dot{C} \cap i_{p_\alpha, g}(S') \neq \emptyset\}$ is dense below q .

Proof.

- 1' Let $r_0 \leq q$ be arbitrary and let us find $r \leq r_0$ with $r \in D$.
- 2' α_{n+1}, r_{n+1} are chosen recursively such that:
 - a. $r_{n+1} < r_n$
 - b. $r_n \cap \text{Ord} < \alpha_{n+1} < \omega_1^{r_{n+1}}$
 - c. $r_{n+1} \Vdash \alpha_{n+1} \in \dot{C}$.
- 3' Let $\mathcal{M}$ be the curious sequence given by $(r_i : i < \omega)$, with $(j^i : i < \omega)$ the corresponding embeddings. We have that $\mathcal{M}$ is iterable.
- 4' Let $\sigma : p_\alpha \prec p_\alpha^*$ the iteration of p_α given by r_0 .
- 5' Let G be $\mathcal{M}$ -generic such that $G \ni j^0(\sigma)(S')$.
- 6' Let $\mathcal{M}' := \text{Ult}(\mathcal{M}, G)$, with $(\pi^i : i < \omega)$ the ultrapower embeddings. We have $\omega_1^{\mathcal{M}} \in \pi^0(j^0(\sigma))(S')$ and $\mathcal{M}'$ is iterable.
- 7' Let (N, J) be a pre-condition containing a real coding $\mathcal{M}'$. Let ρ be an ω_1 -iteration of $\mathcal{M}'$ inside N satisfying $I^{\mathcal{N}_i} = J \cap \mathcal{N}_i$ for every i .
- 8' Let $a := (\rho^1 \circ \pi^1)(j^0)(a^{r_0})$ and let $r := (N, \in, J, a) \in \mathbb{P}_{\max}$.
- 9' $(\rho^{i+1} \circ \pi^{i+1})(j^i) : r_i \prec (\mathcal{N}_i, \in, I^{\mathcal{N}_i}, a)$ is an iteration belonging to N . Since $I^{\mathcal{N}_i} = J \cap \mathcal{N}_i$, we conclude $r < r_i$.
- 10' By 6°6', $\kappa := \omega_1^{r_0} \in \rho^0(\pi^0(j^0(\sigma)))(S')$.
- 11' Since $\rho^0(\pi^0(j^0(\sigma)))$ is the iteration of p_α given by r , we conclude $r \Vdash \kappa \in i_{p_\alpha, g}(S')$.
- 12' By 6°2', we first have $\kappa = \sup\{\alpha_n : 0 < n < \omega\}$, and consequently $r \Vdash \kappa \in \dot{C}$.
- 13' Thus, $r \Vdash \kappa \in \dot{C} \cap i_{p_\alpha, g}(S')$, so we have $r \in D$ and $r < r_0$.

□

7° It follows from the claim that $g \cap D \neq \emptyset$, which implies $S \cap C = i_{p_\alpha, g}(S') \cap \dot{C}^g \neq \emptyset$.

□

VII. Theorem. Assume $V = L(\mathbb{R}) \models \text{AD}$ and $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Then $V[g] \models \text{ZFC}$.

Proof.

- 1° By 4.3-X and 5-IV, we have $V[g] \models \text{Universe} = L(P(\omega_1)) \wedge \psi_{\text{AC}}$.
- 2° By 5-V, it suffices to show that there is a partition of ω_1 into ω_1 disjoint stationary sets.
- 3° Let $p \in g$ be arbitrary and inside p , let $f : \omega_1 \rightarrow P(\omega_1) - I^p$ satisfy:
 - a. $f(\alpha) \cap f(\beta) = \emptyset$ for $\alpha \neq \beta$;
 - b. $\omega_1 = \bigcup \text{im } f$.

(Such an f exists since $p \models \text{ZFC}$.)

4° Let $i_{p,g} : p \prec p^*$ be the iteration of p given by g . By elementarity, we have $h := i_{p,g}(f) : \omega_1 \rightarrow P^{p^*}(\omega_1) - I^{p^*}$ and it satisfies:

- a. $h(\alpha) \cap h(\beta) = \emptyset$ for $\alpha \neq \beta$;
- b. $\omega_1 = \bigcup \text{im } h$.

5° By 5-VI, we have $P^{p^*}(\omega_1) - I^{p^*} \subseteq \text{NS}_{\omega_1}^+$. Thus, h is a required family.

□

VIII. **Lemma.** Assume that every real has the dagger. Let J be a normal ideal on ω_1 . Let

$$D \subseteq \prod_{p \in \mathbb{P}_{\max}} (P(\omega_1)^p - I^p)$$

be such that for every $p_0 \in \mathbb{P}_{\max}$ and every $d_0 \in P(\omega_1)^{p_0} - I^{p_0}$, there exists $(p, d) \in D$ satisfying $p \leq p_0$ and $d \subseteq i_{p_0,p}(d_0)$. Then for every $p \in \mathbb{P}_{\max}$, there exist an iteration $i : p \prec p^*$, sequence $(p_\alpha : \alpha < \omega_1)$ of conditions, family of iterations $i_{\alpha,\beta} : p_\alpha \prec p_\beta^\beta$ ($\alpha \leq \beta < \omega_1$), and a sequence $(d_\alpha : \alpha < \omega_1)$ such that:

- a. $p_0 = p$ and $p_\beta < p_\alpha$ for $\alpha < \beta$;
- b. $i_{\alpha,\beta}$ is the iteration of p_α given by p_β (for $\alpha \leq \beta < \omega_1$);
- c. $i_{0,\alpha}$ is an initial segment of i ;
- d. $(p_{\alpha+1}, d_\alpha) \in D$ for $\alpha < \omega_1$;
- e. $\nabla \{i_{\alpha+1,\omega_1}(d_\alpha) : \alpha < \omega_1\}$ contains a club;
- f. $I^{p^*} = J \cap p^*$.

Proof.

- 1° This is a variation on the theme of 4.3-VI. We will define sequence $(p_\alpha : \alpha < \omega_1)$ by recursion on α . We also fix enumerations $\{B_i^\alpha : i < \omega\} = P(\omega_1)^{p_\alpha} - I^{p_\alpha}$ as we go.
- 2° Let $(A_\alpha : \alpha < \omega_1)$ be pairwise disjoint J -positive sets whose diagonal union is $\omega_1 - \{0\}$. Let $g : \omega_1 \times \omega \rightarrow \omega_1$ be a bijection satisfying $g(\alpha, i) \geq \alpha$ for every α .
- 3° At stage $\alpha = 0$, let $p_0 := p$.
- 4° Consider recursively stage $\alpha + 1$. Let β, j be such that $\alpha = g(\beta, j)$. There are $p_{\alpha+1} < p_\alpha$ and $d_\alpha \subseteq i_{\beta,\alpha+1}(B_j^\beta)$ such that $(p_{\alpha+1}, d_\alpha) \in D$.
- 5° Let us now consider limit stage γ .
- 6° Let $h_\gamma : \omega \rightarrow \gamma$ be an increasing cofinal function and let $\mathcal{M}_\gamma$ be the curious sequence given by $(p_{h(i)} : i < \omega)$, with $j_\gamma^i : p_{h(i)} \prec (\mathcal{M}_\gamma)_i$ the corresponding iterations.
- 7° By 2-VII, there is a pre-condition (M, I) containing a real coding $\mathcal{M}_\gamma$ such that $M \models \text{MA}_{\omega_1}$.

8° $\omega_1^{\mathcal{M}_\gamma}$ belongs to exactly one A_α . Let us choose an $\mathcal{M}_\gamma$ -generic G_γ satisfying: if $\alpha < \gamma$ and $t < \omega$ is such that $h_\gamma(t) > \alpha$, then $j_\gamma^t(i_{\alpha, h_\gamma(t)}(d_\alpha)) \in G_\gamma$. (Note that here we are simply requiring that G_γ contains one set, when the conditions are right.) Let $\mathcal{M}' := \text{Ult}(\mathcal{M}_\gamma, G_\gamma)$.

9° By 3.2-VI, $\mathcal{M}_\gamma$ is iterable. Consequently, $\mathcal{M}'$ is also iterable.

10° By 3.1-XVI, inside M , there is an ω_1 -iteration $k : \mathcal{M}' \prec \mathcal{N}$ such that $I^{\mathcal{N}_i} = I \cap \mathcal{N}_i$ for every $i < \omega$.

11° Let $a_\gamma := \cup\{a^{p_\alpha} : \alpha < \gamma\}$ and let $p_\gamma := (M, \in, I, k^0(j_\gamma^0(a_\gamma)))$. Clearly, $p_\gamma \in \mathbb{P}_{\max}$ and $p_\gamma < p_\alpha$ for every $\alpha < \gamma$.

12° This concludes the recursion. Iterations i_{α_0} are determined by commutative systems

$$(i_{\alpha, \beta} \upharpoonright p_{\alpha_0}^\alpha : \alpha_0 \leq \alpha \leq \beta < \omega_1)$$

of iterations. We set $i := i_0$.

13° **Claim.** For $\alpha < \omega_1$, we have that $A_\alpha - i_\alpha(d_\alpha)$ is non-stationary.

Proof.

1' $C := \{\omega_1^{\mathcal{M}_\gamma} : \gamma = \cup \gamma \in (\alpha, \omega_1)\}$ is club.

2' For $\gamma = \cup \gamma \in (\alpha, \omega_1)$, if $\omega_1^{\mathcal{M}_\gamma} \in A_\alpha$, then $\omega_1^{\mathcal{M}_\gamma} \in i_\alpha(d_\alpha)$ (cf. 8°).

3' Thus, $C \cap (A_\alpha - i_\alpha(d_\alpha)) = \emptyset$, yielding the conclusion. $\square$

14° Since $\nabla_{\alpha < \omega_1} A_\alpha = \omega_1 - \{0\}$, we conclude that $\nabla_{\alpha < \omega_1} i_\alpha(d_\alpha)$ contains a club.

15° **Claim.** $I^{p^*} = J \cap p^*$

Proof.

1' By 2-IX, we have $I^{p^*} \subseteq \text{NS}_{\omega_1} \subseteq J$.

2' Let us verify $(\supseteq)$. Let $X \in P(\omega_1)^{p^*} - I^{p^*}$ be arbitrary and let us show $X \in P(\omega_1) - J$.

3' We have that $X = i_\alpha(B_s^\alpha)$ for some $\alpha < \omega_1$ and some $s < \omega$. Let $\beta := g(\alpha, s)$

4' By 4°, we have $i_\beta(d_\beta) \subseteq i_\alpha(B_s^\alpha) = X$.

5' By 13°, we have a club C such that $A_\alpha \cap C \subseteq i_\beta(d_\beta) \subseteq X$, yielding that X is J -positive. $\square$

$\square$

IX. **Theorem.** Assume $V = L(\mathbb{R}) \models \text{AD}$ and let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$. Then inside $V[g]$, we have that NS_{ω_1} is saturated.

Proof.

- 1° Let $\tau \in V^{\mathbb{P}_{\max}}$ be such that $\Vdash \ulcorner \tau \subseteq \text{NS}_{\omega_1}^+ \text{ is dense open} \urcorner$. We want to show that $\Vdash \ulcorner \text{there is } A \subseteq \tau \text{ such that } |A| = \omega_1 \text{ and } \nabla A \text{ contains a club} \urcorner$.
- 2° This is done by a density argument. Let $p \in \mathbb{P}_{\max}$ be arbitrary and let us find $p' < p$ forcing the required statement.
- 3° Let us fix the standard (partial) recursive coding $f : \mathbb{R} \rightarrow H_{\omega_1}$. Let A denote the following set:

$$\{x \in \mathbb{R} : (\exists q \in \mathbb{P}_{\max})(\exists e \in P(\omega_1)^q)(q \Vdash i_{q,G}(e) \in \tau \wedge f(x) \downarrow = (q, e))\}.$$

- 4° Let (M, I) be an A -iterable pre-condition containing a real coding p and satisfying

$$(H_{\omega_1}^M, \in, A \cap M) \prec (H_{\omega_1}, \in, A).$$

- 5° We reason inside M . By 5-VIII applied to

$$D := \{(q, e) : (\exists x \in A \cap M)(f(x) \downarrow = (q, e))\},$$

there exist an iteration $i : p \prec p^*$, sequence $(p_\alpha : \alpha < \omega_1)$ of conditions, iterations $i_{\alpha, \beta} : p_\alpha \prec p_\beta$ ($\alpha \leq \beta < \omega_1$), and a sequence $(d_\alpha : \alpha < \omega_1)$ such that:

- a. $p_0 = p$ and $p_\beta < p_\alpha$ for $\alpha < \beta$;
- b. $i_{\alpha, \beta}$ is the iteration of p_α given by p_β (for $\alpha \leq \beta < \omega_1$);
- c. $i_{0, \alpha}$ is an initial segment of i ;
- d. $(p_{\alpha+1}, d_\alpha) \in D$ for $\alpha < \omega_1$;
- e. $\nabla \{i_{\alpha+1, \omega_1}(d_\alpha) : \alpha < \omega_1\}$ contains a club;
- f. $I^{p^*} = J \cap p^*$.

- 6° Let $p' := (M, \in, I, i(a^p)) \in \mathbb{P}_{\max}$. We have $p' \leq p$.
- 7° Let $E := \{i_{\alpha+1, \omega_1^M}(d_\alpha) : \alpha < \omega_1^M\}$.
- 8° By 5°e, we have $p' \models \ulcorner \nabla E \text{ belongs to the filter dual to } I^{p'} \urcorner$. By elementarity and 2-IX, we conclude $p' \Vdash \ulcorner \nabla i_{p', G}(E) \text{ contains a club} \urcorner$.
- 9° **Claim.** $p' \Vdash \nabla i_{p', g}(E) \subseteq \tau$.

Proof.

- 1' Let $g \rightsquigarrow V^{\mathbb{P}_{\max}}$ with $g \ni p'$ and let $(e_\alpha : \alpha < \omega_1) := i_{p', g}((d_\alpha : \alpha < \omega_1^M))$. We want to show that for every $\alpha < \omega_1$, we have $e_\alpha \in \tau^g$.
- 2' Let $(q_\alpha : \alpha < \omega_1) = i_{p', g}((p_\alpha : \alpha < \omega_1^M))$.
- 3' We have $p' \models (\forall \alpha < \omega_1)(\exists x \in A \cap M)(f(x) \downarrow = (p_{\alpha+1}, d_\alpha))$.
- 4' By elementarity and A -iterability, for every $\alpha < \omega_1$, there is $x \in A$ such that $f(x) \downarrow = (q_{\alpha+1}, e_\alpha)$.
- 5' In other words, for every $\alpha < \omega_1$, we have $q_{\alpha+1} \Vdash e_\alpha \in \tau$.
- 6' By 4.3-VII, we have $\{q_\alpha : \alpha < \omega_1\} \subseteq g$, yielding $\{e_\alpha : \alpha < \omega_1\} \subseteq \tau^g$.

□

- 10° Thus, p' is as required.

□

References

- [Jackson, 2010] Jackson, S. (2010). Structural consequences of ad. In *Handbook of set theory*, pages 1753–1876. Springer.
- [Jech, 2003] Jech, T. (2003). *Set Theory*. Springer, Berlin, Heidelberg.
- [Kakuda, 1981] Kakuda, Y. (1981). On a condition for cohen extensions which preserve precipitous ideals. *The Journal of Symbolic Logic*, 46(2):296–300.
- [Kanamori, 2008] Kanamori, A. (2008). *The higher infinite: large cardinals in set theory from their beginnings*. Springer Science & Business Media.
- [Kechris, 1984] Kechris, A. S. (1984). The axiom of determinacy implies dependent choices in $L(\mathbb{R})$. *The Journal of symbolic logic*, 49(1):161–173.
- [Kunen, 2014] Kunen, K. (2014). *Set theory an introduction to independence proofs*. Elsevier.
- [Larson, 2010] Larson, P. B. (2010). Forcing over models of determinacy. In *Handbook of set theory*, pages 2121–2177. Springer.
- [Martin and Steel, 1983] Martin, D. A. and Steel, J. R. (1983). The extent of scales in $L(\mathbb{R})$. In *Cabal Seminar 79–81*, pages 86–96. Springer.
- [Woodin, 2013] Woodin, W. H. (2013). *The axiom of determinacy, forcing axioms, and the nonstationary ideal*. de Gruyter.